Attention and Regret*

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Abstract

This paper develops an optimality foundation for regret theory. We ask which emotional responses to decision outcomes would best incentivize an agent to pay more attention. Regret emerges as the uniquely optimal emotion. Our approach microfounds original regret theory (Bell 1982; Loomes and Sugden 1982) and enables its extension to account for decision complexity. Regret is stronger in simpler decision problems, consistent with a self-blame component. A scaling property allows extrapolation to new complexity environments without additional parameters.

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[Regret] may also depend on the extent to which the individual blames himself for his original decision. [...] The neglect of this dimension of regret—although a useful simplifying assumption for many problems—is a serious obstacle to the development and generalisation of regret theory. - Sugden 1986

I. Introduction

Decisions under uncertainty can trigger regret or rejoicing when one learns that another action would have yielded better or worse outcomes. These emotions are often viewed as futile "crying over spilled milk," and their anticipation has been shown to distort choices in ways that violate expected utility theory (Bell 1982; Loomes and Sugden 1982; Zeelenberg 1999a). This raises a puzzle: evolutionary fitness depends only on realized outcomes, yet we possess emotions triggered by foregone ones. Why, then, do regret and rejoicing exist?

We propose that regret and rejoicing exist to motivate attention, thereby improving decision quality. By attention, we primarily mean information processing—such as deliberation or memory search—though it can also be interpreted as information acquisition. The anticipation of regret and rejoicing increases the incentive to pay attention: a better decision not only improves the material outcome but also avoids regret and brings rejoicing. This effect of regret on attention is empirically supported by Reb (2008), who finds that making regret more salient leads subjects to take longer to reach a decision and to acquire more information. Regret can thus be seen as a penalty for not thinking hard enough, in line with regret having a self-blame component (Connolly and Zeelenberg 2002).

Our functional explanation has a natural interpretation based on parental socialization. When parents consistently blame or praise children for certain outcomes, these evaluations become internalized over time as self-blame or self-congratulation (Kochanska and Thompson 1997; Lagattuta and Thompson 2007). Just as parents instill guilt in this way to discourage norm transgressions and immediate gratification (Tracy, Robins, and Tangney 2007), we propose they instill regret to encourage thinking harder about decisions. This view aligns with the economics literature on intergenerational preference transmission, which emphasizes that parents actively shape children's preferences and values (Bisin and Verdier 2011; Doepke, Sorrenti, and Zilibotti 2019). A central theme in this literature is conflict over patience between parents and children. This conflict translates naturally to attention: since attention is costly and its benefits materialize only later, parents prefer more attention if they are more patient than their children or if children exhibit present bias.

Is regret the optimal emotion for motivating attention? The answer is not obvious a priori because any emotion rewarding good outcomes and penalizing bad ones could motivate attention by raising the stakes of the decision. Using a mechanism-design approach, we show that regret and rejoicing indeed emerge as uniquely optimal. The resulting model takes precisely the form of original regret theory (Bell 1982; Loomes and Sugden 1982), including its key functional form assumption, while extending regret theory to account for decision complexity.

Our model draws on the principal-agent approach to the evolution of preferences (e.g., Robson 2001b; Rayo and Becker 2007). A parent (principal) shapes a child's (agent's) emotional responses to incentivize attention. From the parent's perspective, the child is biased toward insufficient attention due to disagreement in time preferences. Through rewards and penalties that become internalized, the parent shapes the child's emotional utility as a function of the chosen action and realized state. Two observability constraints shape the problem: the parent observes neither how much attention the child pays, inducing a hidden-action problem, nor the child's exact bias, inducing a hidden-information problem. The first constraint captures that attention is internal to the child's mind, while decision outcomes are observable. The second reflects that emotions are socialized in early childhood, when preferences are malleable but not yet stable; the parent cannot target the exact bias their child will eventually have. We model attention using the rational-inattention paradigm, where the agent can choose any signal structure subject to an entropy-based information cost (Sims 2003). This creates a high-dimensional mechanism-design problem with both hidden information and hidden actions. To render the model tractable, we restrict to decision problems that are invariant under relabeling actions and impose that emotional utility preserves this symmetry.

Regret and rejoicing are optimal because they amplify utility differences between actions depending on the material stakes in each state. Formally, let $v(a, \omega)$ denote the material utility under action a and state ω. In our model, regret or rejoicing experienced in state ω under action a is a function of the vector of material utility differences $v(a, \omega) - v(a', \omega)$ to alternative actions a' under the realized state ω.[1] By rewarding better actions and penalizing worse ones, regret and rejoicing amplify the statewise utility differences between actions, which we call the *stakes* in a given state. The emotional amplification of stakes motivates the agent to pay more attention to the state. Because regret and rejoicing involve comparisons to alternative actions, they amplify stakes by an amount that depends on the material utility differences between actions, that is, the material stakes. This is optimal because material stakes determine how much attention a state deserves. By depending on material stakes, these emotions optimally direct attention to states where stakes are higher.

For binary decision problems which are the canonical setting in regret theory, we obtain sharper results. The optimal emotions yield *original regret theory* (Bell 1982; Loomes and Sugden 1982), including its key assumption, disproportionate aversion to large regrets. This assumption means that regret amplifies larger utility differences between actions disproportionately. Disproportionate aversion to large regrets explains well-documented violations of expected utility theory, including the Allais paradox, simultaneous gambling and insurance, and preference reversals (Bell 1982; Loomes and Sugden 1982). Our model predicts a weaker version of this property—starshapedness rather than convexity—but we demonstrate that starshapedness is sufficient to generate the above violations of expected utility that motivated the convexity assumption.

[1] This contrasts with disappointment and elation, which also involve counterfactual comparisons but of a different kind. While regret involves *between-prospects* comparisons—what would have happened under a different action—disappointment involves *within-prospect* comparisons—what would have happened under a different state.

In our model, disproportionate aversion to large regrets arises from heterogeneity in the agent's bias toward insufficient attention. When the material stakes are already large, agents with small biases choose the ex-post optimal action most of the time, and their mistake rates are relatively insensitive to further increases in the stakes. In contrast, agents with stronger biases make more frequent mistakes, and their behavior is more responsive to stake increases. The principal therefore tailors the emotional amplification to the more biased types by scaling up larger material stakes disproportionately more.

Original regret theory has two limitations: it does not account for decision complexity and was developed only for binary choice. A key benefit of our microfoundation is that it addresses both while disciplining how these extensions should be made.

First, our model predicts that regret depends on the complexity of the decision problem. As emphasized by Sugden (1986) and decision-justification theory (Connolly and Zeelenberg 2002), regret has a self-blame component, and self-blame depends on how justifiable a decision was. A mistake is less justifiable if the decision problem was easy, so simpler decision problems should elicit more regret following mistakes. Our model features a complexity parameter λ that captures how costly it is to discover the right action, and indeed predicts that simpler problems elicit stronger regret. Our model also predicts that regret-based distortions to choice and attention vanish as decision problems become very simple or very complex.

Extending regret theory to incorporate decision complexity raises concerns about introducing new parameters. One might expect a different regret function for each complexity level, requiring separate measurement for each, which would limit the model's applicability and falsifiability. A virtue of our microfoundation is that it disciplines this extension. A scaling property implies that regret at any complexity can be derived from regret at a reference complexity via a simple transformation, so no additional preference parameters are needed. The complexity parameter can be identified from the stochasticity of choice, following standard methods in rational inattention (Caplin 2016).

Second, our model predicts regret beyond binary choice, which has proven difficult for original regret theory (Loomes and Sugden 1982; Loomes and Sugden 1987). A frequently used model for general choice sets considers only regret relative to the ex-post best action (Quiggin 1994). Our model makes the psychologically plausible prediction that regret and rejoicing are affected by all counterfactual actions, in line with generalized regret theory (Loomes and Sugden 1987; Sugden 1993). Our model strengthens generalized regret theory by predicting that regret depends only on material utility differences and that regret maintains the utility ordering of actions.

Our results have testable implications for feedback interventions. Regret is mediated by counterfactual feedback—whether decision-makers learn ex post what would have happened under alternative choices (Zeelenberg 1999a). Previous work has shown that the expectation of counterfactual feedback induces choice distortions in line with regret theory (Zeelenberg 1999a; Filiz-Ozbay and Ozbay 2007). Our paper emphasizes that counterfactual feedback also affects attention. Providing feedback increases attention by inducing regret but distorts attention toward high-stakes states.

Moreover, these effects depend on decision complexity. The incentive effects of regret are strongest for simpler problems (Proposition 3), while attention and choice distortions peak at medium complexity (Proposition 4). These results can inform the optimal use of feedback interventions, which are cheap ways to incentivize attention but generate predictable distortions.

Related Literature We contribute to the large literature on regret in economics, starting with Bell (1982) and Loomes and Sugden (1982). We introduce their model, original regret theory, in section A.1. Bleichrodt and Wakker (2015) review empirical support and applications of regret theory across domains such as finance, health, insurance, and auctions. We provide an optimality foundation for original regret theory and simultaneously extend the theory to incorporate decision complexity.

Our approach follows the shift in emotions research toward functional accounts of emotions (Keltner and Gross 1999), and shares with cultural constructionist accounts an emphasis on cultural influences on emotions (Mesquita and Parkinson 2024). Psychologists often think of regret as an adaptive emotion that improves decision-making (e.g., Janis and Mann 1977; Zeelenberg 1999b). This view is supported by Camille et al. (2004), who find that patients with damage to their orbitofrontal cortex, who do not experience regret but are otherwise cognitively intact, make worse decisions. We provide an attention-based mechanism through which regret improves decisions.[2] The attention mechanism is empirically supported by Reb (2008), who finds that making regret salient leads subjects to take longer to reach decisions and to acquire more information.

A popular alternative hypothesis for how regret may benefit decision-making is that it improves learning in repeated decisions. When agents experience regret, they purportedly learn to modify their future behavior. This hypothesis draws theoretical support from reinforcement learning with regret, called regret matching, which has desirable asymptotic properties (Foster and Vohra 1999).[3] However, regret in reinforcement learning and regret in decision theory are very different. Regret matching predicts stochastic behavior where the probability of an action is proportional to its historical regret. By contrast, regret theory predicts agents choose the action with the highest expected utility, where utility has a regret component. Our framework provides an optimality foundation that delivers the behavioral predictions of regret theory (see Section III.A.3).

This paper relates to the economics literature on intergenerational preference transmission (Bisin and Verdier 2011; Doepke, Sorrenti, and Zilibotti 2019). Our approach follows models where parents actively shape children's preferences (Bisin and Verdier 2001; Doepke and Zilibotti 2017), rather than models of cultural transmission through imitation (Boyd and Richerson 1985). Whereas previous work typically assumes parents choose a preference parameter, we focus on a microfounded technology of preference transmission that takes observability constraints seriously: parents reward or penalize observable outcomes, and this parental feedback becomes internalized

[2] Compte (2025) develops a model where the agent influences an internal regulator (or second self) through psychosomatic costs, with an application to regret as a way to motivate the regulator's attention. His model restricts to negative emotions, assumes emotional costs are borne by both selves, and allows for asymmetric decision problems.

[3] Regret also appears prominently in reinforcement learning as a performance benchmark for learning algorithms ("no regret learning"). Regret matching is one of many learning algorithms that achieves this benchmark.

as experienced utility. This technology allows parents to flexibly shape children's utility functions within the observability constraint, connecting to the principal-agent approach to the evolution of preferences (Robson 2001b; Rayo and Becker 2007).

At a broader level, our paper contributes to the literature on how a principal should incentivize an expert to acquire information, beginning with Demski and Sappington (1987). Our model allows for flexible information acquisition, as in recent contributions such as Lindbeck and Weibull (2020) and Yoder (2022), and is distinctive in combining this with both hidden actions (signal structures are not contractible) and hidden information (the agent's bias is private). We contribute to this literature by showing that counterfactual comparisons (as they factor into regret) are useful for incentivizing information acquisition.

Yoder (2022) also considers flexible information acquisition and hidden information, but assumes that either the signal structure or the signal realization is contractible. In contrast, in our model only the action and realized state are contractible. We show that self-screening is ineffective and hence our solution methods and the resulting distortions differ: in his model, there is no distortion at the top, whereas in our case, information acquisition is skewed towards extreme states for all types. Methodologically, Yoder (2022) uses the posterior approach (Caplin, Dean, and Leahy 2022), while we use the state-dependent choice probabilities approach (Matějka and McKay 2015) and its connection to perturbed utility models (Bloedel, Denti, and Pomatto 2025).

The paper is organized as follows. Section II introduces the model. Section III presents the results for binary and larger choice sets, and the extension incorporating decision complexity. Section IV concludes. The Appendix contains proofs and discusses alternative model interpretations. The Online Appendix shows how to extend the model to incomplete ex-post feedback about the state of the world, where disappointment and elation arise to supplement regret and rejoicing to motivate attention.

II. Model

A principal (she) shapes the preferences of an agent (he) who faces a decision problem under uncertainty. The agent can pay attention to the state of the world before choosing an action, but is biased toward insufficient attention from the principal's perspective. The principal instills emotions as a function of the chosen action and the realized state. Our primary interpretation is cultural evolutionary: parents (principal) shape children's (agent) emotional responses through evaluations that become internalized over time.

Decision Problem A *decision problem* is a tuple (Ω, μ, A, v). It comprises a finite state space Ω, a full-support prior $\mu \in \Delta(\Omega)$ on the state space, a finite set of actions A, and a material utility function $v\colon A \times \Omega \to \mathbb{R}$.[4]

[4] We assume a finite state space to circumvent measure-theoretic issues. We believe our results extend to infinite state spaces under appropriate technical assumptions.

Attention The agent faces a two-stage problem. First, he acquires a signal structure about the state (attention stage); second, he chooses an action from A conditional on the realized signal (decision-making stage). A signal structure is a pair (S, σ), where S is a finite signal space and $\sigma \colon \Omega \to \Delta(S)$ is a state-dependent distribution over signals, with $\sigma(s|\omega)$ denoting the probability of signal s in state ω. Following the rational-inattention approach (Sims 2003), the agent can select any signal structure subject to the mutual information cost[5]

$$c(S, \sigma) := \sum_{s \in S} \Big(- \sigma(s) \log \sigma(s) \Big) - \sum_{\omega \in \Omega} \left(\mu(\omega) \sum_{s \in S} \Big(- \sigma(s|\omega) \log \sigma(s|\omega) \Big) \right),$$

where $\sigma(s) := \sum_{\omega \in \Omega} \mu(\omega)\sigma(s|\omega)$ is the unconditional probability of signal s and log refers to the natural logarithm (with $0 \log 0 := 0$). The first term is the entropy of the unconditional signal distribution; the second is the expected entropy of the signal conditional on the state. Entropy captures the uncertainty of a distribution, so the cost reflects the reduction in uncertainty about the signal from knowing the state—that is, how much the signal is tailored to the state.

As is well known, it suffices to consider signal structures that recommend an action, i.e., $S = A$. The reason is that mutual information is Blackwell monotonic, so the agent benefits from garbling any signal structure into an action recommendation (Matějka and McKay 2015). This reduces the agent's two-stage problem to selecting state-dependent choice probabilities $p \colon \Omega \to \Delta(A)$, written $p(a|\omega)$, which we call an *attention strategy*. Writing $c(p)$ for $c(A, p)$, the attention cost captures how much the action is tailored to the state.

Principal's Preferences The principal derives utility $V(p)$ from attention strategy p:

$$V(p) = \sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p(a|\omega) v(a, \omega) - \lambda c(p).$$

The principal internalizes the attention cost weighted by λ and internalizes the agent's material utility $v(a, \omega)$ from action a under state ω. We interpret λ as the complexity of the decision problem, since it scales the cost of learning about the state.

Agent's Bias The agent has a bias $\beta \in (0, 1]$ that discounts the material utility $v(a, \omega)$, or, equivalently, overweights the attention cost by $1/\beta$. Under our interpretation, this captures that children pay less attention than parents want them to. Parents have many reasons to want children to pay more attention than children would themselves—for instance, because parents may not fully internalize the attention costs borne by children, or because parents have a direct stake in children's decision outcomes. We focus on disagreement about patience. Since attention is costly and its

[5] While we assume the canonical mutual information cost, which has foundations in information theory (Shannon 1948), we expect most of our results to hold for the broader class of Csiszár information costs (Bloedel, Denti, and Pomatto 2025). These costs satisfy the state-separability property that is crucial for our main results.

benefits realize only later, $\beta \leq 1$ reflects that children are less patient than parents.[6] Disagreements about patience are central to the literature on intergenerational preference transmission, which has studied how parents socialize children to value future-oriented investments such as hard work and education (Lindbeck and Nyberg 2006; Doepke and Zilibotti 2017).

The disagreement between parents and children can arise from two sources. First, parents are more patient than children and have paternalistic preferences: they use their own preferences to evaluate children's choices (Bisin and Verdier 2001; Doepke and Zilibotti 2017). Second, the child is tempted to shirk on attention due to present bias, and parents want to correct for this bias. Present bias is one of the most studied biases in behavioral economics, with substantial evidence for its prevalence (Imai, Rutter, and Camerer 2021). Under this interpretation, regret serves as an internal self-control mechanism for paying sufficient attention.

The bias β is heterogeneous and private information of the agent, with cumulative distribution function F. We make two assumptions on the distribution of the bias β. First, the bias is bounded away from zero, which ensures that finite emotions are optimal. Second, $\log \beta$ admits a strictly log-concave density, which is used to show uniqueness.[7]

Assumption 1. *The bias β has support $\operatorname{supp}(\beta) = [\underline{\beta}, 1]$ where $\underline{\beta} > 0$. The distribution of $\log \beta$ admits a strictly log-concave density.*

Heterogeneity of preferences and biases across individuals is well documented. Under the interpretation that parents are more patient than children and impose their own time preferences (paternalism), heterogeneity in β reflects heterogeneity in children's patience—more impatient children have a larger bias from the parent's perspective. Studies consistently find substantial variation in time preferences within populations (Frederick, Loewenstein, and O'Donoghue 2002). Under the interpretation that parents correct for children's present bias, heterogeneity in β arises because present bias varies across individuals (Augenblick and Rabin 2019).

The bias is private information of the agent. Emotions are socialized in childhood (Eisenberg, Cumberland, and Spinrad 1998; O'Connor, McCormack, and Feeney 2012), when preferences are malleable but not yet stable. Because emotional responses must be instilled before preferences have fully formed, parents cannot target the specific β their child will eventually have. The distribution of β captures this residual uncertainty about the child's realized bias. Under the present-bias interpretation, an additional source of uncertainty arises: self-control varies from situation to situation. Cognitive load reduces self-control (Shiv and Fedorikhin 1999), as do stress and cues that trigger visceral urges (Loewenstein 1996). Thus, even if a parent could perfectly predict their child's baseline patience, the effective bias in any given decision remains stochastic.[8]

[6] The assumption $\beta \leq 1$ can be replaced by the weaker condition $\mathbb{E}[\beta^2] \leq \mathbb{E}[\beta]$, which allows for some probability mass above 1, see footnote 19.

[7] Both assumptions are satisfied by common families truncated to $[\underline{\beta}, 1]$, including Beta(a, b) distributions with $b > 1$ and log-normal distributions.

[8] There is also a strategic rationale for why tailoring emotions to β is difficult even if β is permanent. Lemma 1 shows that if parents conditioned emotions on the child's reported bias, this would not benefit parents: higher types prefer stronger incentives (due to their higher return to attention) and select into them, while lower types select

Emotions The principal shapes the agent's preferences over decision outcomes by choosing an emotion-inclusive utility function $u : A \times \Omega \to \mathbb{R}$. We interpret $m(a, \omega) := u(a, \omega) - v(a, \omega)$ as emotional utility, which can depend on the chosen action and the realized state. The agent with bias β derives utility U_β from the attention strategy p,

$$U_\beta(p) = \beta \sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p(a|\omega) u(a, \omega) - \lambda c(p).$$

The underlying technology is that parental evaluations of children's observed behavior become internalized over time. When parents consistently blame or praise children in response to certain outcomes, children learn to anticipate these reactions, which eventually become self-generated emotional responses. This mechanism is central to the development of conscience and self-conscious emotions, where parental reactions become internalized as self-evaluations (Kochanska and Thompson 1997; Lagattuta and Thompson 2007).

Crucially, emotions can only condition on observable behavior—the chosen action and the realized state—not on how much attention was paid. This reflects that attention is largely internal to the child's mind, while decision outcomes are observable.[9] This constraint is crucial for the optimality of regret: the principal cannot directly reward attention, only outcomes.[10]

Our parenting technology rules out that parents can directly modify the child's patience—that is, the weight on decision outcomes relative to attention costs. While parents may also attempt to teach patience, the persistence of present bias and parent-child conflicts over patience reveals limits to this strategy—possibly because teaching patience is not enforced through observables, unlike praise and blame for decision outcomes. Our model focuses on inculcating emotions as a function of observable behavior, taking the child's bias β as given—reflecting whatever teaching may have accomplished. Moreover, emotions dominate teaching patience unless parents can tailor patience to the child's private type β. To see this, note that any uniform increase in β by a factor k can be replicated by scaling up utilities through emotions: $u(a, \omega) = k \cdot v(a, \omega)$. But optimal regret is non-linear and state-dependent (Theorem 1), achieving what uniform patience increases cannot.

Socialization Cost as Tie-Breaker The model so far identifies optimal emotions only up to statewise constants. The reason is that the principal cares only about the agent's behavior, and shifting all emotional utilities in a given state by a constant does not alter behavior because the

weaker incentives. This is the opposite of what the principal wants, so screening backfires. The same adverse-selection logic applies if parents use the children's behavior to infer β and tailor emotions. Children would have incentives to manipulate their behavior to obtain their preferred incentive scheme, limiting the scope for tailoring emotions.

[9] As a concrete example, consider a child deciding where or how to play. Some ways of playing are riskier: running near a pool, climbing on furniture. The state captures whether the situation is safe, such as whether the ground is wet, the structure is stable. By paying attention (observing the environment, recalling past experiences, thinking through what could go wrong) the child can learn about the state. The parent observes the child's choice and whether the situation was safe, but not how carefully the child thought before acting. Parents can then blame or praise based on these observables.

[10] If the principal could condition on attention, she could achieve the first best by penalizing deviations from the optimal attention strategy. Similarly, if the bias were observable, she could achieve the first best by scaling up utilities by $1/\beta$.

agent is an expected-utility maximizer.[11] For readers interested only in behavioral predictions, this indeterminacy is immaterial. For the interpretation of $m = u - v$ as experienced emotions, statewise constants determine whether emotions are positive or negative. To resolve this indeterminacy, we introduce a socialization cost (Bisin and Verdier 2001; Doepke and Zilibotti 2017).

In case of indifference, the principal selects the utility function u that minimizes the expected socialization cost

$$\Phi(u) := \int \left(\sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p_\beta(a|\omega) \phi\big(u(a,\omega) - v(a,\omega)\big) \right) dF(\beta), \tag{1}$$

where $u(a,\omega) - v(a,\omega) = m(a,\omega)$ is the internalized emotion and $\phi(m)$ is the socialization cost of making children internalize this emotion. Both positive and negative emotions are costly to socialize. We assume ϕ is strictly convex, differentiable, and minimized at 0.

Assumption 2. *The socialization cost $\phi\colon \mathbb{R} \to \mathbb{R}$ is strictly convex, differentiable, and minimized at 0.*

This tie-breaker can be understood as the limit of a model where the expected socialization cost Φ enters the principal's objective additively with vanishing weight relative to $V(p)$. If socialization costs entered the objective, the optimal emotion would still have the structure of regret, but the shape of regret—its magnitude and sensitivity to stakes—would be affected by the socialization cost ϕ. We adopt the limiting case to ensure that emotions are shaped by the optimal incentives for attention rather than the functional form of ϕ.

The principal's objective is material utility v, not emotion-inclusive utility u. Following the emotion-socialization literature (Hoffman 1983; Kochanska 1993), we treat emotions as tools to shape behavior rather than as sources of well-being to be maximized. This is consistent with the prominent role of negative emotions—guilt, shame, regret—in emotional development, which suggests that parents do not strongly internalize children's emotional experiences. This assumption can be relaxed. Suppose parents partially internalize children's emotional well-being, so the effective cost becomes $\tilde{\phi}(m) = \phi(m) - \alpha m$ for some $\alpha > 0$. If $\phi(m) = |m|$ and $\alpha < 1$, then $\tilde{\phi}$ is still minimized at zero, leaving the analysis unchanged. For differentiable ϕ, the cost $\tilde{\phi}$ is minimized at some $a > 0$ rather than at zero. This is equivalent to a model with some cost function $\hat{\phi}$ satisfying our assumptions, combined with a uniform upward shift in all emotions by a. The structure of optimal emotions is preserved; only the level changes.

Symmetry The presence of both hidden actions and hidden information creates a high-dimensional mechanism-design problem. In particular, the agent's attention strategy generally does not have a closed-form solution. To render the problem tractable, we make two symmetry assumptions.

[11] As in regret theory, *given emotions*, our agent is an expected-utility maximizer. However, because emotions can be a function of the choice set, the agent's choice behavior can violate expected-utility axioms. To be precise, that statewise constants do not affect behavior also relies on the information cost being additively separable, so there are no "income effects" on attention from better emotions.

First, we impose that the material utility v is symmetric in the actions as defined below. Second, we restrict ourselves to emotions that maintain this symmetry. Under symmetric utility, the agent's attention strategy takes the multinomial logit form (Matějka and McKay 2015); without symmetry, the solution involves action biases that depend on the entire decision problem and generally lack closed-form expressions.

Definition 1. *A function $w\colon A \times \Omega \to \mathbb{R}$ is **symmetric** if the distribution over vectors $(w(a,\omega))_{a\in A} \in \mathbb{R}^A$ induced by prior μ is invariant under permutation of the coordinates.*

Symmetric utility means that the decision problem is invariant under relabeling the actions. As an example, the utility function according to the payoff matrix in Table 1 is symmetric if $\mu(\omega_1) = \mu(\omega_2)$ and $\mu(\omega_3) = \mu(\omega_4)$. That is because the utility vector $(0,1)$ is as likely as its permutation $(1,0)$ and the utility vector $(3,2)$ is as likely as $(2,3)$.

	ω_1	ω_2	ω_3	ω_4
a_1	1	0	3	2
a_2	0	1	2	3

Table 1: Symmetric payoff matrix

Assumption 3. *The material utility v is symmetric.*

This assumption is plausible when the actions are indistinguishable before information processing (Matějka and McKay 2015), a natural benchmark for situations where all actions are "plausible candidates for choice" (Sugden 1986) and the agent has no strong prior intuition. This describes typical applications of regret theory in finance, insurance, and health, where the decision-maker chooses among assets, insurance contracts, or medical procedures without an ex-ante favorite.

For tractability, we focus on emotion-inclusive utility functions u that maintain symmetry. Thus, the emotions do not depend on the labeling of actions, but we otherwise allow for a large non-parametric class of emotions, including disappointment and elation (Loomes and Sugden 1986). This symmetry assumption is natural because the underlying decision problem is symmetric in the actions, so the first-best attention strategy is symmetric in the actions.[12]

[12] Simulations under two states suggest that inducing asymmetries through emotions does not benefit the principal. For a quadratic posterior-separable information cost under two states, one can show that asymmetric utility is suboptimal. The symmetry restrictions on v and u can be dropped if one replaces the rational-inattention model with a perturbed-utility model (Fudenberg, Iijima and Strzalecki 2015), in which the agent pays a cost to control their behavior. Under an entropic control cost, the agent's choice probabilities follow multinomial logit even for asymmetric decision problems, so the analysis extends.

Principal's Problem Combining the above, the principal chooses an emotion-inclusive utility function $u\colon A\times\Omega\to\mathbb{R}$ and attention strategies $p_\beta\colon\Omega\to\Delta(A)$ to solve the following problem (P).

$$\max_{u,(p_\beta)_\beta}\int\left(\sum_{\omega\in\Omega}\sum_{a\in A}\mu(\omega)p_\beta(a|\omega)v(a,\omega)-\lambda c(p_\beta)\right)dF(\beta) \tag{P}$$

$$\text{s.t.}$$

$$\forall\beta\colon\ \ p_\beta\in\arg\max_{p}\ \beta\sum_{\omega\in\Omega}\sum_{a\in A}\mu(\omega)p(a|\omega)u(a,\omega)-\lambda c(p) \tag{IC}$$

$$u \text{ is symmetric.} \tag{SYM}$$

Out of the solutions to (P), the principal chooses the one that minimizes the socialization cost $\Phi(u)$.

One may wonder whether the principal could tailor emotions to the agent's type by asking the agent to report their bias. However, such screening is typically valuable only when the principal wants different types to take different actions. Here, the principal wants the agent to pay the same amount of attention regardless of the agent's bias. In Appendix A, we formally show that under two states, screening cannot improve upon inculcating a single emotion, so the formulation of (P) is without loss. In fact, any attempt to screen backfires: higher types select stronger incentives while lower types select weaker incentives, increasing the variance of attention.

III. Results

First, we characterize the solution of (P) for decision problems containing two actions and then turn to the case of $n>2$ actions. Assumptions 1 to 3 are maintained throughout.

A. Two Actions

This section considers binary decision problems, $A=\{a_1,a_2\}$. We first introduce original regret theory, then characterize the optimal emotion in our model, and finally compare our solution to existing regret theories.

A.1 Original Regret Theory

To verify that the optimal emotion of our model can be interpreted as regret and rejoicing, we introduce *original regret theory* due to Bell (1982) and Loomes and Sugden (1982).

The idea behind original regret theory is that upon learning the realized state, agents experience not only realized payoff but also emotional utility depending on the payoff difference to the counterfactual action. When choosing between two actions, agents compare expected emotion-inclusive utility. Formally, let $\mathcal{A}$ be a grand set of actions. Original regret theory assumes a material utility function $v:\mathcal{A}\times\Omega\to\mathbb{R}$, a regret-rejoicing function $R:\mathbb{R}\to\mathbb{R}$, and a subjective belief $\pi\in\Delta(\Omega)$.

Action a_i is weakly preferred to action a_j, written $a_i \succsim a_j$, if and only if

$$\sum_{\omega\in\Omega} \pi(\omega)\Big(v(a_i,\omega) + R\big(v(a_i,\omega) - v(a_j,\omega)\big)\Big) \geq \sum_{\omega\in\Omega} \pi(\omega)\Big(v(a_j,\omega) + R\big(v(a_j,\omega) - v(a_i,\omega)\big)\Big). \quad (2)$$

Here, $R(x)$ is the emotional utility from a material utility difference x to the counterfactual action: $R(x) < 0$ if $x < 0$ (regret) and $R(x) > 0$ if $x > 0$ (rejoicing).

While intuitive, the representation in terms of R contains redundancy: different functions R can induce identical preferences. The literature therefore typically works with the following equivalent formulation. Subtracting the right-hand side from the left-hand side of (2), we obtain

$$a_i \succsim a_j \Leftrightarrow \sum_{\omega\in\Omega} \pi(\omega) Q\big(v(a_i,\omega) - v(a_j,\omega)\big) \geq 0, \quad (3)$$

where $Q(x) := x + R(x) - R(-x)$ is the emotion-inclusive utility difference between actions in a state where the material utility difference is x. By construction, Q is skew-symmetric, $Q(-x) = -Q(x)$.

Equation (3) shows that Q captures all behavioral consequences of regret. Original regret theory is defined as preferences of the form (3) with skew-symmetric Q. Diecidue and Somasundaram (2017) axiomatize this representation, and Bleichrodt, Cillo, and Diecidue (2010) show how to identify v and Q from choices.

Under non-linear Q, regret theory allows for violations of expected utility theory such as non-transitive preferences. Bell (1982) and Loomes and Sugden (1982) observe that if Q is strictly convex, regret theory explains the Allais paradox, the common-ratio effect, and simultaneous gambling and insurance.

A.2 Optimal Emotion

We now define what it means for the solution to the principal's problem to take the form of regret theory. Our definition follows original regret theory but requires only that Q is starshaped rather than convex.

Definition 2. *A function $g : \mathbb{R} \to \mathbb{R}$ is **starshaped** if $g(0) = 0$, $g(x) > 0$ for $x > 0$, and if for all $x > 0$ and $k > 1$:*

$$g(kx) > k\, g(x).$$

This property is also known as strictly increasing returns to scale in the context of production functions. Our definition of starshapedness lies between strict convexity and strict superadditivity for nonnegative functions on $[0,\infty) \to \mathbb{R}$ that vanish at 0 (Bruckner and Ostrow 1962).

For a_i, $i = 1, 2$, denote by a_{-i} the other action in A, that is, $a_{-1} = a_2$ and $a_{-2} = a_1$.

Definition 3. *Under two actions, $A = \{a_1, a_2\}$, a solution $(u, (p_\beta)_\beta)$ to the principal's problem* (P) *is a **regret solution** if there exists a continuous function $R\colon \mathbb{R} \to \mathbb{R}$ such that for all $(a_i, \omega) \in A \times \Omega$:*

$$u(a_i,\omega) = v(a_i,\omega) + R\big(v(a_i,\omega) - v(a_{-i},\omega)\big)$$

and

1. $R(x) \gtreqless 0 \Leftrightarrow x \gtreqless 0$,

2. $Q(x) := x + R(x) - R(-x)$ *is starshaped.*

Note that a regret solution requires not only that the agent's preferences can be represented as in regret theory, but that the emotional utility m itself takes the form of the regret-rejoicing function in Bell (1982) and Loomes and Sugden (1982).

Property 1 states that emotional utility is positive (rejoicing), zero, or negative (regret) according to whether the counterfactual material utility is worse, equal to, or better than the actual material utility. The requirement that R, and thus Q, is continuous is important for measurement (Diecidue and Somasundaram 2017). Property 2 requires that regret and rejoicing increase more than proportionally as a function of the material utility difference, which we also refer to as disproportionate aversion to large regrets. Section III.A.3 shows that starshaped Q delivers the behavioral predictions of regret theory.

The following theorem shows that the unique solution to the principal's problem is a regret solution. This is perhaps surprising: one might expect the optimal emotion to depend on the entire decision problem, that is, all material utilities and the prior. Instead, emotional utility depends only on the material utility difference between the chosen and counterfactual action in the realized state. Moreover, the regret-rejoicing function R is independent of the decision problem, though it depends on λ and the distribution of β, which we use for comparative statics below.

Theorem 1. *Under a binary decision problem, the unique solution to the principal's problem* (P) *is a regret solution, where R is independent of the decision problem.*

The theorem states that optimal emotions take the form of original regret theory. The intuition for why a regret solution is optimal is as follows.

Recall that the principal seeks to motivate more attention from the agent. Define the *emotion-inclusive stakes* $\Delta u(\omega) := u(a_1, \omega) - u(a_2, \omega)$ as the emotion-inclusive utility difference between actions. The key observation is that these stakes determine how much attention the agent pays to a state. Under rational inattention, the agent's choice probability is given by a logit function (Matějka and McKay 2015):

$$p_\beta(a_1|\omega) = \frac{e^{\frac{\beta}{\lambda}\Delta u(\omega)}}{1 + e^{\frac{\beta}{\lambda}\Delta u(\omega)}} \tag{4}$$

Higher emotion-inclusive stakes lead to a higher probability of choosing the correct action. Regret and rejoicing motivate attention by increasing these stakes. Property 1 reflects this: regret penalizes the wrong action and rejoicing rewards the right action.

The preceding explains why emotions raise the stakes. But why do they depend on counterfactual comparisons—the difference to what one could have gotten? Define the *material stakes* as $\Delta v(\omega) := v(a_1, \omega) - v(a_2, \omega)$. The material stakes capture the value of getting the decision right in a state and thus determine how much attention the principal wants the agent to pay. Optimal emotions are a function of the material stakes, to align the agent's attention with what is optimal.

The map from material to emotion-inclusive stakes is therefore the key object the principal chooses. This map is Q: given material stakes Δv, the better action yields rejoicing $R(\Delta v)$ whereas the worse action yields regret $R(-\Delta v)$, so emotion-inclusive stakes are $\Delta v + R(\Delta v) - R(-\Delta v) = Q(\Delta v)$. We now turn to the shape of Q.

Starshapedness of Q stems from uncertainty about the bias β. If β were known, the optimal Q would be linear, scaling up the stakes by the inverse of the bias:

$$Q(\Delta v) = \frac{\Delta v}{\beta}. \tag{5}$$

This would perfectly offset the agent's bias, implementing the first-best attention strategy with no distortions to attention or choices. Under unknown β, the optimal Q is characterized by the implicit equation:

$$Q(\Delta v) = \frac{\Delta v}{\frac{\int \rho'_\beta(Q(\Delta v))\beta \, dF(\beta)}{\int \rho'_\beta(Q(\Delta v)) \, dF(\beta)}}, \tag{6}$$

where $\rho_\beta(x) = e^{\frac{\beta}{\lambda}x}/(1 + e^{\frac{\beta}{\lambda}x})$ denotes here the logit choice probability as a function of the stakes. The optimal Q scales up stakes by the inverse of a weighted average of the bias. The weights are the sensitivities ρ'_β of the choice probabilities to the stakes. For higher material stakes, less biased types (high β) already choose correctly with high probability, so their choice probabilities are insensitive to further stake increases. The more biased types remain responsive. Thus, at higher stakes, the weighting shifts toward more biased types, who require larger emotional amplification. Put differently, mistakes in high-stakes states occur only for strongly biased agents and thus warrant larger penalties relative to the material stakes. This yields disproportionate aversion to large regrets: Q is starshaped rather than linear.

We now turn to the behavioral implications of optimal regret, beginning with attention. Choice distortions are addressed below. While regret's effects on choice are well studied, its implications for attention have been overlooked. We show that regret skews attention toward extreme states.

To state this formally, we measure the *extremeness* of a state ω by the material stakes

$$|\Delta v(\omega)| := |v(a_1, \omega) - v(a_2, \omega)|.$$

We measure *attention to state* ω by the log-likelihood ratio of of the signal (equivalently, the action) conditional on that state. By equation (4), the type-β agent's log-likelihood ratio is

$$\ell_\beta(\omega) := \log\left(\frac{p_\beta(a_1|\omega)}{p_\beta(a_2|\omega)}\right) = \frac{\beta}{\lambda} Q\big(\Delta v(\omega)\big).$$

The first-best attention strategy p^* yields log-likelihood ratio $\ell^*(\omega) = \frac{1}{\lambda}\Delta v(\omega)$.

Corollary 1. *The ratio $\ell_\beta(\omega)/\ell^*(\omega)$ is strictly increasing in the extremeness of state ω: relative to* *the first-best, the agent pays disproportionately more attention to more extreme states.*

The intuition is as follows. The agent's attention to a state is determined by the emotion-inclusive stakes. Disproportionate aversion to large regrets—the starshapedness of Q—amplifies stakes more in states where material stakes are already high, skewing attention toward extreme states. Formally, the ratio $\ell_\beta(\omega)/\ell^*(\omega)$ equals $\beta Q(x)/x$ where $x = \Delta v(\omega)$, and starshapedness together with skew-symmetry imply that $Q(x)/x$ is increasing in $|x|$.

The prediction of disproportionate attention to high-stakes states is similar in spirit to models of salience or focusing based on contrast (Bordalo, Gennaioli and Shleifer 2012; Kőszegi and Szeidl 2013). Salience effects are normally understood as a feature of automatic bottom-up attention rather than deliberate top-down attention such as rational inattention (Bordalo, Gennaioli and Shleifer 2022). We show that such focusing effects are not only consistent with rational inattention, but are in fact optimal distortions to motivate attention.[13]

A.3 Behavioral Implications for Choice under Uncertainty

This section examines the behavioral predictions of our model for choice under uncertainty. We show that the starshapedness of Q delivers the empirical violations of expected utility theory associated with regret theory. At first glance, starshapedness may appear to sacrifice predictive power: convexity assumptions are central to both original regret theory and the subsequently developed generalized regret theory (Loomes and Sugden 1987). However, we find that starshaped Q is sufficient. First, starshaped Q implies a generalized regret representation, so our model inherits all predictions of generalized regret theory, including preference reversals of the type documented by Grether and Plott (1979). Second, in the context of original regret theory, starshapedness is equivalent to the conjunction of the Allais paradox, the common-ratio effect, and simultaneous gambling and insurance—the behavioral anomalies that motivated the convex Q assumption of original regret theory. Our microfoundation thus delivers precisely the functional form needed to generate regret theory's behavioral predictions.

We first define the relevant preference representations, following Herweg and Müller (2021). Let $\mathcal{A}$ denote a set of actions.

Definition 4. *A preference relation $\succsim$ over $\mathcal{A}$ has an* ***original regret representation*** *if there exist $\pi \in \Delta(\Omega)$, $v : \mathcal{A} \times \Omega \to \mathbb{R}$, and a skew-symmetric, continuous function $Q : \mathbb{R} \to \mathbb{R}$ such that for all $a_i, a_j \in \mathcal{A}$,* (3) *holds.*

Original regret theory typically assumes that Q is strictly increasing and strictly convex on $\mathbb{R}_{>0}$. Our model predicts the weaker property that Q is starshaped.

To compare our predictions to generalized regret theory, we introduce monetary outcomes. Let $x : \mathcal{A} \times \Omega \to \mathbb{R}$ denote the monetary outcome of action a under state ω. For the rest of this section, we assume the material utility is a strictly increasing, continuous function of monetary outcomes:

[13] Lieder, Griffiths, and Hsu (2018) offer an alternative explanation for overrepresentation of extreme events based on the resource-rationality approach.

$v(a,\omega) = \bar{v}(x(a,\omega))$ for some strictly increasing, continuous $\bar{v} : \mathbb{R} \to \mathbb{R}$. Generalized regret theory (Loomes and Sugden 1987) works directly with monetary outcomes rather than utilities, assuming a skew-symmetric function $\Psi : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ over pairs of monetary outcomes.

Definition 5. *A preference relation $\succsim$ over $\mathcal{A}$ has a* ***generalized regret representation*** *if there exist $\pi \in \Delta(\Omega)$ and a skew-symmetric, continuous function $\Psi : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ such that for all $a_i, a_j \in \mathcal{A}$,*

$$a_i \succsim a_j \Leftrightarrow \sum_{\omega \in \Omega} \pi(\omega)\Psi(x(a_i,\omega), x(a_j,\omega)) \geq 0. \tag{7}$$

Original regret theory is the special case where $\Psi(x,y) = Q(\bar{v}(x) - \bar{v}(y))$. Fishburn (1989) and Sugden (1993) axiomatize generalized regret theory. Loomes and Sugden (1987) impose two additional properties on Ψ:[14]

- **Increasing**: for all $x, y, z \in \mathbb{R}$, $x \gtreqless y \Leftrightarrow \Psi(x,z) \gtreqless \Psi(y,z)$.
- **Convex**: for all $x > y > z$, $\Psi(x,z) > \Psi(x,y) + \Psi(y,z)$.

Under these properties, generalized regret theory shares many predictions with original regret theory (Loomes and Sugden 1987).

Proposition 1. *If preferences have an original regret representation with starshaped Q, they have a generalized regret representation with increasing and convex Ψ.*

Despite the suggestive name, convexity of Ψ does not require convexity of Q—contrary to claims in the literature (e.g., Quiggin 1990, p. 505). The proof is straightforward. Define $\Psi(x,y) := Q(\bar{v}(x) - \bar{v}(y))$. Then Ψ inherits skew-symmetry and continuity from Q, so preferences have a generalized regret representation. For convexity, the inequality $\Psi(x,z) > \Psi(x,y) + \Psi(y,z)$ reduces to $Q(a+b) > Q(a) + Q(b)$ where $a := \bar{v}(x) - \bar{v}(y)$ and $b := \bar{v}(y) - \bar{v}(z)$. This is superadditivity, which follows from starshapedness.

Proposition 1 establishes that our model inherits all predictions of generalized regret theory. However, original regret theory with convex Q may generate additional predictions. Does weakening convexity to starshapedness sacrifice these? We show that it does not: starshapedness is equivalent to the Allais paradox, the common-ratio effect, and simultaneous gambling and insurance—the phenomena that motivated the convexity assumption on Q.

The following definitions, from Loomes and Sugden (1982), formalize these phenomena. Given a belief π over states, an action induces a lottery over monetary outcomes, allowing us to identify actions with lotteries. Two independent lotteries are identified with two actions such that the distributions over monetary outcomes induced by the prior are independent. Let $(x_1, p; x_2, 1-p)$ denote a lottery giving monetary outcome x_1 with probability p and x_2 otherwise, and let (x, p) be shorthand for $(x, p; 0, 1-p)$.

[14] Loomes and Sugden (1987) also impose an ordering property, but it is implied by increasingness and skew-symmetry.

Definition 6 (Allais Paradox). *Let $X_1 = (x_1, p_1; x_2, \alpha)$ and $X_2 = (x_2, p_2 + \alpha)$ be independent lotteries where $1 \geq p_2 > p_1 > 0$ and $(1 - p_2) \geq \alpha \geq 0$. If there exists some probability $\bar{\alpha}$ such that $X_1 \sim X_2$ when $\alpha = \bar{\alpha}$, then (i) if $x_1 > x_2 > 0$, then $\alpha < \bar{\alpha} \Rightarrow X_1 \succ X_2$ and $\alpha > \bar{\alpha} \Rightarrow X_1 \prec X_2$, and (ii) if $0 > x_2 > x_1$, then $\alpha < \bar{\alpha} \Rightarrow X_1 \prec X_2$ and $\alpha > \bar{\alpha} \Rightarrow X_1 \succ X_2$.*

Definition 7 (Common-Ratio Effect). *Let $X_1 = (x_1, \lambda p)$ and $X_2 = (x_2, p)$ be independent lotteries, where $1 \geq p > 0$ and $1 > \lambda > 0$. If there exists some probability $\bar{p}$ such that $X_1 \sim X_2$ when $p = \bar{p}$, then (i) if $x_1 > x_2 > 0$, then $p < \bar{p} \Rightarrow X_1 \succ X_2$ and $p > \bar{p} \Rightarrow X_1 \prec X_2$, and (ii) if $0 > x_2 > x_1$, then $p < \bar{p} \Rightarrow X_1 \prec X_2$ and $p > \bar{p} \Rightarrow X_1 \succ X_2$.*

Definition 8 (Simultaneous Gambling and Insurance). *Let $\bar{v}$ be linear in money. Let $X_1 = (0, 1)$ and $X_2 = (x, p; -px/(1-p), 1-p)$ be independent lotteries where $0 < p < 1$ and $x > 0$. Then (i) for $p > 0.5$, $X_1 \succ X_2$, (ii) for $p < 0.5$, $X_1 \prec X_2$, and (iii) for $p = 0.5$, $X_1 \sim X_2$.*

We now show that these three phenomena are characterized by starshapedness.

Proposition 2. *Suppose preferences have an original regret representation. The following are equivalent: (i) preferences exhibit the Allais paradox, the common-ratio effect, and simultaneous gambling and insurance; (ii) Q is starshaped.*

The key observation is that the arguments in Loomes and Sugden (1982) require weaker properties than convexity: the Allais paradox and common-ratio effect follow from superadditivity of Q, while simultaneous gambling and insurance is equivalent to starshapedness. Thus, our model predicts the behavioral anomalies that motivated original regret theory, without requiring the stronger convexity assumption.

Finally, we address a methodological point. In our model, beliefs arise endogenously from attention, whereas regret theory, following the Savage framework, takes beliefs as given. But conditional on any belief, the agent's preferences over actions in our model are exactly as in regret theory. Given the equilibrium beliefs that agents acquire, the Appendix F shows that choices are undistorted relative to expected material utility—the agent chooses the same action as a material-utility maximizer. It is a striking result that regret solves the attention problem so effectively that its choice distortions do not materialize in equilibrium. Given other beliefs—for example, resulting from trembles in attention or exogenous information arrival—choices would be distorted exactly as regret theory predicts.

Regret theory has been applied to choices between lotteries, where the agent's belief is assumed to match the objective probabilities. We follow the standard evolutionary methodology—applicable to both biological and cultural evolution—of deriving preferences for representative environments and applying them more broadly (Robson 2001a; Rayo and Becker 2007; Netzer 2009).[15] We take

[15] These papers derive optimal utility functions for deterministic choice problems, then interpret the resulting preferences in risky-choice contexts—a larger leap than in our model, where the agent already faces uncertainty at the decision-making stage. More recently, Netzer, Robson, Steiner, and Kocourek (2025) extend the approach of Robson (2001a) and Netzer (2009) directly to risky choice, though they still assume preferences apply beyond the environment for which they evolved.

the representative environment to be one where the agent can learn about the state. This is natural in most real-world settings, including the typical applications of regret theory: investors can research assets, patients can learn about treatments, bidders can investigate auctioned objects. Our representative environment excludes decisions between lotteries, where the agent faces explicit objective probabilities and hence cannot learn about the state. But such environments are rare outside economic experiments. Thus, we assume that when the agent faces a lottery, the preferences derived for the representative environment still apply, generating the behavioral anomalies characterized in Proposition 2.

B. More than Two Actions

Regret theory does not easily generalize to multiple actions because regret induces nontransitive preferences in pairwise choice. Several models have been proposed. Quiggin (1994) assumes that regret depends on the difference between the actual payoff and the payoff of the ex-post optimal action. Loomes and Sugden (1987) propose generalized regret theory, where utility depends on the realized outcome and the outcomes of all counterfactual actions under the same state. Our solution is closest to the latter but imposes additional structure. We define a *multiaction regret solution* as follows.

Definition 9. *Under a finite choice set A, a solution $(u, (p_\beta)\beta)$ to the principal's problem* (P) *is a* ***multiaction regret solution*** *if there exists a function R such that for all $(a, \omega) \in A \times \Omega$:*

$$u(a, \omega) = v(a, \omega) + R\big(\{v(a, \omega) - v(a', \omega)\}_{a' \in A \setminus \{a\}}\big),$$

and for all $a, a' \in A$, $\omega \in \Omega$:

$$v(a, \omega) > v(a', \omega) \Leftrightarrow u(a, \omega) > u(a', \omega).$$

According to a multiaction regret solution, emotions are a state-independent function of the set of material utility differences to counterfactual actions, so a permutation of counterfactual material utilities does not change the emotion, nor does shifting all material utilities in a state by a constant. Emotions preserve the strict ordering of utilities within each state, as in Sarver (2008). Our representation has more content than generalized regret theory: regret depends on material utility differences rather than outcomes. This parallels how original regret theory strengthens generalized regret theory for two actions. Our representation is more general than Quiggin (1994): regret and rejoicing are affected by all counterfactual actions, not just the ex-post best one. This is psychologically plausible—one may regret not having chosen an action even if it were not ex-post optimal, and rejoicing necessarily involves comparison to ex-post suboptimal alternatives.[16]

[16] Quiggin (1994) shows irrelevance of statewise dominated alternatives (ISDA) together with generalized regret theory implies regret depends only on the ex-post best action. In our context of modeling attention, one must distinguish between objective and subjective statewise domination. An action is objectively statewise dominated if some other action yields higher material utility in every objectively possible state. But recognizing domination may

Finally, for technical reasons we assume that the principal has strict preferences over actions in each state. This can be interpreted as a genericity assumption or as the state being sufficiently fine-grained.

Assumption 4. *For all $\omega \in \Omega$, for all $a, a' \in A$: $(a \neq a' \to v(a,\omega) \neq v(a',\omega))$.*

Under these assumptions, we prove the following theorem.

Theorem 2. *For any finite A, the principal's problem* (P) *has a multiaction regret solution, where R is independent of the decision problem.*

The theorem shows that optimal emotions take the form of a multiaction regret solution. Moreover, given λ and the distribution of β, the model can be used to compute the precise optimal emotional utilities. While optimal emotions could depend on the whole decision problem, they reduce to a statewise $|A|$-variable optimization problem: optimal emotions are a function of the vector of material utilities in the realized state. They are given by the socialization-cost-minimizing solution to

$$\max_{u(\cdot,\omega):\, A \to \mathbb{R}} \int \left(\sum_{a \in A} \Big(p_\beta(a|u(\omega)) v(a,\omega) \Big) + \lambda H\big(p_\beta(u(\omega))\big) \right) dF(\beta), \tag{8}$$

where $p_\beta(a|u(\omega))$ is given by the multinomial logit formula

$$p_\beta(a|u(\omega)) = \frac{e^{\frac{\beta}{\lambda} u(a,\omega)}}{\sum_{a' \in A} e^{\frac{\beta}{\lambda} u(a',\omega)}},$$

and $H(p_\beta(u(\omega)))$ is the entropy of the probability vector $p_\beta(u(\omega)) = (p_\beta(a|u(\omega)))_{a \in A}$. Unfortunately, the problem does not admit a simple analytical solution or comparative statics as in binary decision problems. However, once solved for a vector of material utilities, the solution applies to payoff-equivalent states in any decision problem.

C. Decision Complexity

As emphasized by Sugden (1986) and decision-justification theory (Connolly and Zeelenberg 2002), regret has a self-blame component. Taking self-blame seriously, regret theory should account for whether one could have easily known that another action would turn out better: a mistake in an easy decision is more blameworthy than the same mistake in a complex one. Thus, regret should

require information processing: a health insurance plan may be objectively statewise dominated, but this may not be clear before paying attention. An action is subjectively statewise dominated if some other action yields higher material utility in every state the agent considers possible before information processing. Such actions are ruled out by our symmetry assumption on v, according to which all actions are equally serious candidates for choice ex ante. If the model were extended to include subjectively dominated alternatives, one would expect the principal to ensure they do not affect choices, satisfying subjective ISDA. Actions that are objectively but not subjectively statewise dominated are consistent with our symmetry assumption. For such actions, ISDA loses its normative appeal: the intuition that one should ignore dominated alternatives presumes one knows they are dominated. Nevertheless, simulations suggest that violations of objective ISDA in our model are small: the optimal solution is close to Quiggin's model. Specifically, if a^* is the best action in state ω, then $\forall a : u(a,\omega) - u(a^*,\omega) \approx Q(v(a,\omega) - v(a^*,\omega))$ where Q is as in binary decision problems. Proving this remains an open problem.

depend on the complexity of a decision problem. Incorporating decision complexity into regret theory raises two challenges: how exactly complexity should be operationalized, and the concern that a separate regret function would be needed for each complexity level, proliferating the model's degrees of freedom. Our model addresses both challenges.

Our approach operationalizes complexity through the information cost parameter λ, which scales the cost of learning about the state, following the information-acquisition literature (e.g., Gonçalves 2024).[17] Some decisions are inherently simpler than others: in medical decisions, the optimal treatment is easier to identify for some conditions; in auctions, the value of some goods is more easily discernible. We now let λ vary across decision problems and index the optimal emotions accordingly, writing R_λ and Q_λ.

The following results characterize how decision complexity affects optimal emotions in the case of two actions.

Proposition 3. *Under two actions:*

1. *(Scaling) For any $\lambda, \lambda' > 0$ and $x > 0$:*

$$Q_{\lambda'}(x) = \frac{\lambda'}{\lambda} Q_\lambda \left(\frac{\lambda}{\lambda'} x \right).$$

2. *(Comparative Statics) For $\lambda' < \lambda$: $Q_{\lambda'}(x) > Q_\lambda(x)$ and $|R_{\lambda'}(-x)| > |R_\lambda(-x)|$ for all $x > 0$.*

The scaling property disciplines the extension of regret theory to incorporate complexity. Measuring Q at a single reference complexity suffices to extrapolate the model to any other complexity level, so no additional preference parameters are needed. The reason lies not in our specific attention cost but in additive separability: under any additively separable attention cost, λ enters the agent's choice probabilities only through the ratio $\beta u/\lambda$, so scaling material utilities and complexity by the same factor scales the optimal emotion-inclusive utility by the same factor. The complexity parameter itself can be identified from state-dependent stochastic choice data (Caplin 2016): more complex problems induce choice probabilities closer to uniform.

The comparative statics show that simpler problems (lower λ) induce larger Q_λ. A larger Q_λ means that regret and rejoicing amplify material stakes more, providing stronger incentives for attention. Thus, the incentive effects of regret are strongest in simple decision problems. This is relevant for feedback interventions, which induces regret by providing counterfactual feedback: the attentional benefits are larger in simpler problems. Mathematically, the result follows from the scaling property and starshapedness. The intuition is similar to that for starshapedness: when the right action is easy to discover, only agents with strong biases make mistakes, and these agents require larger emotional amplification to offset their bias.

[17] This operationalization of complexity is closely related to two other popular approaches: noise in choices (Enke and Shubatt 2023) and choice confidence (Enke and Graeber 2023). A higher information cost parameter λ leads to more noise in state-contingent choice and less confidence that one's decision is ex-post correct, as the agent acquires less precise information about the state.

Simpler problems also induce stronger regret, $|R_{\lambda'}(-x)| > |R_\lambda(-x)|$, consistent with the self-blame dimension: mistakes are punished more severely when the right action was easier to discover. Two mechanisms contribute. First, a larger Q means a larger sum of regret and rejoicing. Second, a larger Q increases the probability of taking the right action, so regret is experienced less often than rejoicing, making regret relatively cheaper to induce for the principal.

The next proposition characterizes the limiting behavior of optimal emotions as decision complexity becomes extreme. In both limits, emotion-inclusive utility differences become proportional to material utility differences, so the disproportionate response to larger stakes—which drives violations of expected utility theory—disappears.

Proposition 4 (Limits)**.** *Under any finite action set A, let $(u_\lambda)_{\lambda>0}$ be a selection of solutions to the principal's problem. For all $a, a' \in A$ and $\omega \in \Omega$:*

1. *As $\lambda \to 0$: $u_\lambda(a,\omega) - u_\lambda(a',\omega) \to \big(v(a,\omega) - v(a',\omega)\big)/\underline{\beta}$*
2. *As $\lambda \to \infty$: $u_\lambda(a,\omega) - u_\lambda(a',\omega) \to \big(v(a,\omega) - v(a',\omega)\big) \cdot \mathbb{E}[\beta]/\mathbb{E}[\beta^2]$*

Convergence is uniform over v in any compact set.

The proof of this proposition is in the Online Appendix.

Remark 1. *Under two actions, Proposition 4 is equivalent to:*

1. *As $\lambda \to 0$: $Q_\lambda(x) \to \frac{1}{\underline{\beta}} \cdot x$*
2. *As $\lambda \to \infty$: $Q_\lambda(x) \to \frac{\mathbb{E}[\beta]}{\mathbb{E}[\beta^2]} \cdot x$*

In very simple decision problems ($\lambda \to 0$), the optimal emotion perfectly offsets the bias of the most biased type, $\underline{\beta}$. In very complex problems ($\lambda \to \infty$), the emotion offsets a weighted average of biases, $\mathbb{E}[\beta^2]/\mathbb{E}[\beta]$, which places more weight on higher types. In both limits, Q_λ is linear in stakes, so the starshapedness property that generates the Allais paradox and related anomalies vanishes. Thus, regret-based distortions to choice and attention are strongest in problems of intermediate complexity.

IV. Conclusion

Behavioral models in economics are typically developed based on explanatory power and simplicity. This paper takes a complementary approach based on the *function* of non-expected utility preferences. The usefulness of this approach is illustrated in three ways. First, it provides a criterion to select between different regret theories: our model delivers original regret theory rather than generalized regret theory, and for multiple actions, it predicts that regret depends on all counterfactual actions rather than only the ex-post best one. Second, it extends regret theory to incorporate decision complexity without proliferating degrees of freedom. Third, it generates new testable predictions regarding feedback interventions: providing counterfactual feedback increases attention but distorts it toward high-stakes states, with effects that vary systematically with decision complexity.

Several open questions remain. Our analysis focuses on symmetric decision problems for tractability. Extending the model to asymmetric problems, where some actions are ex-ante more attractive, could reveal whether optimal emotions introduce biases toward certain actions or distort decision-making to incentivize attention. For multiple actions, our model delivers a general regret solution, but whether additional properties can be obtained remains open.

On a broader scale, our research underscores parallels between internal psychological mechanisms, such as counterfactual comparisons, and more traditional principal-agent problems. We believe that such a perspective has the potential to profit both behavioral economics and mechanism design. Behavioral economics might glean insights from mechanism design to better understand the function and benefits of non-standard preferences. Similarly, mechanism design can turn to human psychology as a source of innovative solutions for agency problems, which evolution has already solved.

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Appendix: Proofs

A. Screening is Ineffective

This section analyzes whether the principal could benefit from screening the agent by asking the agent to report their bias and tailoring emotions to the reported bias. We maintain the assumption that any emotion induces an symmetric decision problem for the agent. At the end of the section, we consider a general mechanism. In both cases, the principal does not profit from screening the agent.

The principal chooses emotion-inclusive utility functions $u_\beta\colon A\times\Omega\to\mathbb{R}$ and attention strategies $p_\beta\colon \Omega\to\Delta(A)$, indexed by type, subject to incentive compatibility (including truthful reporting and optimal attention) and symmetry:

$$\max_{(u_\beta,p_\beta)_\beta}\int\left(\sum_{\omega\in\Omega}\sum_{a\in A}\mu(\omega)p_\beta(a|\omega)v(a,\omega)-\lambda c(p_\beta)\right)dF(\beta) \tag{P'}$$

s.t.

$$\forall\beta\colon\ (\beta,p_\beta)\in\arg\max_{\hat\beta,\hat p}\beta\sum_{\omega\in\Omega}\sum_{a\in A}\mu(\omega)\hat p(a|\omega)u_{\hat\beta}(a,\omega)-\lambda c(\hat p) \tag{IC'}$$

$$\forall\beta\colon\ u_\beta \text{ is symmetric.} \tag{SYM'}$$

Out of the solutions, the principal chooses the one minimizing the socialization cost (1).

Lemma 1. *Let* $|\Omega|=2$. *Under Assumptions 1 to 3, for any* $(m_\beta,p_\beta)_\beta$ *satisfying* (IC') *and* (SYM'), *there exists* $(m'_\beta,p'_\beta)_\beta$ *with* $m'_\beta=m'_{\beta'}$ *for all* β,β', *satisfying* (IC') *and* (SYM'), *that achieves a weakly higher payoff for the principal in* (P').

The proof shows that offering a menu makes types sort adversely. The principal would want to offer stronger incentives to more biased types (lower β), but higher types have higher returns to attention and select stronger incentives, contrary to the principal's aim. This increases the variance of attention, decreasing the principal's utility. Thus, the formulation (P) is without loss. The proof holds for any posterior-separable attention cost with strictly concave, differentiable H that is symmetric in the states.

Proof. Under symmetry of v with two states $\Omega=\{\omega_1,\omega_2\}$, either v is statewise constant, or $\mu(\omega_1)=\mu(\omega_2)=\frac{1}{2}$ and there are two actions $A=\{a_1,a_2\}$ satisfying $v(a_1,\omega_1)=v(a_2,\omega_2)$ and $v(a_1,\omega_2)=v(a_2,\omega_1)$. In the former case, zero attention is optimal, and the principal's solution entails $u_\beta=v$ for all β. Consider the other case. Without loss, $v(a_1,\omega_1)=v(a_2,\omega_2)=v>0$ and $v(a_1,\omega_2)=v(a_2,\omega_1)=0$, so the principal's payoff matrix can be written as follows.

	ω_1	ω_2
a_1	v	0
a_2	0	v

Agent's Behavior. By symmetry, any u in the menu $(u_\beta)_\beta$ can be written as

	ω_1	ω_2
a_1	$\underline{u}+\Delta u$	$\underline{u}$
a_2	$\underline{u}$	$\underline{u}+\Delta u$

where $\underline{u}=u(a_1,\omega_2)=u(a_2,\omega_1)$ and $\Delta u=u(a_1,\omega_1)-u(a_2,\omega_1)$.

We characterize the agent's attention strategy given some chosen u by the posterior approach Caplin, Dean, and Leahy (2019). Identifying posteriors $\gamma \in [0,1]$ with the probability of ω_1, type β given u maximizes

$$\max_{\tau \in \Delta[0,1]} \mathbb{E}_\tau[N_\beta(\gamma|u)] \quad \text{s.t.} \quad \mathbb{E}_\tau[\gamma] = \tfrac{1}{2}$$

where the net value function N_β is

$$N_\beta(\gamma|u) = \beta(\underline{u} + \max\{\gamma\Delta u, (1-\gamma)\Delta u\}) + \lambda H(\gamma).$$

The gross value function (first term) is convex and symmetric around $1/2$; the attention cost $H(\gamma) = -\gamma\log(\gamma) - (1-\gamma)\log(1-\gamma)$ is strictly concave and also symmetric around $1/2$. As known from Bayesian persuasion, the solution is obtained by concavifying the net utility function (Kamenica and Gentzkow 2011).

By symmetry, the solution can be characterized by the unique posterior $\gamma \in [1/2, 1]$ that maximizes the net value function. Since the maxima of the net value function are symmetric around the prior $1/2$, there exists a Bayes-consistent τ supported exactly on these maxima. Restricted to $[1/2, 1]$, the net value function is the sum of a linear gross value and the strictly concave attention cost, so there is a unique optimal γ. The first-order condition $-\lambda H'(\gamma) = \beta\Delta u$ and strict concavity of H imply that the optimal γ is strictly increasing in Δu. Given $v > 0$, the principal never offers $\Delta u < 0$, which would induce the agent to pay costly attention only to select the wrong action.

When facing a menu $(u_\beta)_\beta$, the agent chooses a utility function and then an attention strategy. By the above, this is equivalent to finding the posterior γ that maximizes the *effective* value function on $[1/2, 1]$:

$$N_\beta(\gamma|(u_\beta)_\beta) := \max_{\hat\beta}\{N_\beta(\gamma, u_{\hat\beta})\} = \beta\max_{\hat\beta}\{\underline{u}_{\hat\beta} + \gamma\Delta u_{\hat\beta}\} + \lambda H(\gamma) \text{ for } \gamma \in [1/2, 1]$$

By $\Delta u_\beta > 0$ for all β, the effective value function is strictly supermodular in β and γ, so the optimal γ is non-decreasing in β. Since the effective gross value $\beta\max_{\hat\beta}\{\underline{u}_{\hat\beta} + \gamma\Delta u_{\hat\beta}\}$ is the maximum of increasing linear functions in γ, non-decreasing γ implies non-decreasing choice of slope $\Delta u_{\hat\beta}$.

To summarize: (1) higher β implies weakly higher posterior γ; (2) higher β implies weakly higher chosen slope $\Delta u_{\hat\beta}$; (3) given u, the optimal posterior $\gamma \in [1/2, 1]$ is strictly increasing in Δu.

Principal's Problem. The principal's net value function

$$N_P(\gamma) = \max\{\gamma v, (1-\gamma)v\} + \lambda H(\gamma)$$

is strictly (quasi-)concave on $[1/2, 1]$.

Suppose $u_\beta \neq u_{\beta'}$ for some types. Let $\beta^* \in \arg\max_{\beta \in \text{supp}(\beta)} N_P(\gamma_\beta)$. By (2), types above β^* select weakly higher slopes Δu and types below select weakly lower slopes Δu. If the principal offered only u_{β^*} then by (3), posteriors for types below β^* would weakly increase, posteriors for types above would weakly decrease, and γ_{β^*} would be unchanged. By (1), the ordering of posteriors is preserved, so each γ_β moves weakly toward γ_{β^*}. By strict quasi-concavity of N_P and γ_{β^*} being optimal, the principal weakly benefits, and strictly so for any type whose posterior moved strictly. If a positive measure of types has strictly different slopes Δu, the principal strictly benefits by (3). Otherwise, only a measure zero set of types is affected, and the principal is indifferent. In either case, offering a single emotion achieves a weakly higher payoff. □

General Mechanism. A more general mechanism could elicit an additional message after attention. We sketch why this does not help. Given any such mechanism, fix type β and consider the attention problem after the first message. The agent faces an effective value function $\hat{N}_\beta$ given by the maximum over available emotions. Since each emotion induces a symmetric value function, so does the maximum, and the set of maximizers M_β is symmetric around 1/2.

The principal prefers, among solutions to the agent's problem, those with posteriors maximizing N_P. Let $M_{P,\beta} := \arg\max_{\gamma \in M_\beta} N_P(\gamma)$, which is symmetric. Since the prior is 1/2, there exists a Bayes-consistent, binary, symmetric distribution supported on $M_{P,\beta}$.

Selecting this principal-preferred solution for each type: by symmetry, each type picks the same emotion after either posterior. This can be implemented by offering only that emotion after the first message, reducing to the case above.

B. Theorem 2

Proof. Let $A = \{a_1, \ldots, a_n\}$. Because the principal chooses u that is symmetric by (SYM), the (IC) constraint implies that the state-dependent choice probabilities are (Matějka and McKay 2015):

$$p_\beta(a|\omega) = \frac{e^{\frac{\beta}{\lambda} u(a,\omega)}}{\sum_{a' \in A} e^{\frac{\beta}{\lambda} u(a',\omega)}} = \frac{e^{\frac{\beta}{\lambda}(u(a,\omega) - u(a_1,\omega))}}{1 + \sum_{a' \neq a_1} e^{\frac{\beta}{\lambda}(u(a',\omega) - u(a_1,\omega))}}. \tag{9}$$

By (9), the choice probabilities depend only on statewise utility differences with respect to some reference action. Using this, we show that the principal's problem separates by state.

Substituting (9) into (P), the principal chooses only u and the (IC) constraint is satisfied by construction. The resulting problem consists of an outer minimization problem regarding the socialization cost $\phi(u(a,\omega) - v(a,\omega))$ and an inner problem to maximize the material utility net of attention cost, subject to the symmetry constraint:

$$\min_{u\colon A \times \Omega \to \mathbb{R}} \int \left(\sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p_\beta(a|\omega) \phi\big(u(a,\omega) - v(a,\omega)\big) \right) dF(\beta) \tag{10}$$

s.t.

$$u \in \underset{\substack{u\colon A \times \Omega \to \mathbb{R}, \\ u \text{ is symmetric}}}{\arg\max} \int \sum_{\omega \in \Omega} \mu(\omega) \left(\lambda H(p_\beta(\omega)) + \sum_{a \in A} p_\beta(a|\omega) v(a,\omega) \right) dF(\beta) \tag{11}$$

where $p_\beta(a|\omega)$ is given by (9) and $H(p_\beta(\omega)) = \sum_{a \in A} -p_\beta(a|\omega) \log(p_\beta(a|\omega))$ is the entropy of the conditional choice probabilities. We have used that under symmetric utilities, the unconditional choice probabilities are all equal to $1/n$, so the constant entropy of the unconditional choice probabilities can be dropped.

Inner Maximization Problem. We first relax the inner maximization problem by dropping the symmetry constraint. Then, the inner objective is separable by state as $p_\beta(\omega)$ depends only on the vector of statewise utilities, $u(\omega) := (u(a_1, \omega), \ldots, u(a_n, \omega))$. Below, we show that any solution to the relaxed statewise maximization problem

$$\max_{u(\cdot,\omega)\colon A \to \mathbb{R}} \int \left(\lambda H(p_\beta(\omega)) + \sum_{a \in A} p_\beta(a|\omega) v(a, \omega) \right) dF(\beta) \tag{12}$$

preserves the strict order of utilities and that a solution exists. Then, we show that there is a symmetric utility function that maximizes (12) state-by-state. By every state having positive probability, this implies that any solution to the constrained inner-maximization problem solves the relaxed statewise problem (12) in every state.

Lemma 2. *Any solution to the relaxed statewise maximization problem (12) satisfies*

$$\forall a, a' \in A\colon v(a, \omega) > v(a', \omega) \Rightarrow u(a, \omega) > u(a', \omega).$$

Proof. If $v(a, \omega) > v(a', \omega)$ but $u(a, \omega) < u(a', \omega)$, exchanging $u(a, \omega)$ and $u(a', \omega)$ leaves entropy unchanged (by symmetry of H in the probabilities) while improving expected material utility. Hence $u(a, \omega) \geq u(a', \omega)$ at any maximum.

For strict inequality, suppose $v(a, \omega) > v(a', \omega)$ and $u(a, \omega) = u(a', \omega)$. Setting $\tilde{u}(a, \omega) = u(a, \omega) + \varepsilon$ and $\tilde{u}(a', \omega) = u(a', \omega) - \varepsilon$: by symmetry of (9) at $\varepsilon = 0$, we have $p_\beta(a|\omega) = p_\beta(a'|\omega)$, and by symmetry of H, $\frac{dH}{d\varepsilon} = 0$, while $\frac{d}{d\varepsilon} \sum_{\hat{a}} p_\beta(\hat{a}|\omega) v(\hat{a}, \omega) > 0$. Hence $u(a, \omega) > u(a', \omega)$. $\square$

By the genericity Assumption 4, the lemma implies

$$v(a, \omega) > v(a', \omega) \Leftrightarrow u(a, \omega) > u(a', \omega).$$

Before we show existence, we rewrite the statewise objective in (12), which we denote by $G(u(\omega))$, as

$$G(u(\omega)) = \int \left(\sum_{a \in A} p_\beta(a|u(\omega)) \big(v(a, \omega) - \beta u(a, \omega) \big) + \lambda \log \left(\sum_{a \in A} e^{\frac{\beta}{\lambda} u(a, \omega)} \right) \right) dF(\beta). \tag{13}$$

The first term is the difference between the principal's and the agent's utility and the second term is the agent's statewise utility by Caplin, Dean, and Leahy (2019).

It will be useful to know the derivatives of G and $p_\beta(a|\omega)$ with respect to the components of u:

$$\frac{dG(u(\omega))}{du(a,\omega)} = \int \left(\sum_{a' \in A} \frac{dp_\beta(a'|\omega)}{du(a,\omega)} \big(v(a',\omega) - \beta u(a',\omega)\big) \right) dF(\beta) \tag{14}$$

$$\frac{dp_\beta(a|\omega)}{du(a,\omega)} = \frac{\beta}{\lambda} p_\beta(a|\omega)(1 - p_\beta(a|\omega)) \tag{15}$$

$$\frac{dp_\beta(a'|\omega)}{du(a,\omega)} = -\frac{\beta}{\lambda} p_\beta(a'|\omega) p_\beta(a|\omega), \text{ for } a \neq a'. \tag{16}$$

Lemma 3. *The relaxed statewise maximization problem (12) has a solution. Moreover, any solution satisfies*

$$u(a_i,\omega) \leq (i-1)\frac{v(a_n,\omega)}{\inf \beta}$$

for $i = 1, \ldots, n$, assuming $v(a_1,\omega) \leq \cdots \leq v(a_n,\omega)$ and normalizing $u(a_1,\omega) = v(a_1,\omega) = 0$.

Proof. We show that any $u(\cdot,\omega) \in \mathbb{R}^A$ is weakly dominated by some $\tilde{u}(\cdot,\omega) \in \mathbb{R}^A$ in a compact set, on which the maximum is obtained by continuity of the objective.

The normalizations $v(a_1,\omega) = u(a_1,\omega) = 0$ are without loss: shifting all $u(a,\omega)$ by a constant does not affect choice probabilities, and shifting all $v(a,\omega)$ by a constant c changes the objective by exactly c.

By Lemma 2, if $v(a,\omega) > v(a',\omega)$ and $u(a',\omega) > u(a,\omega)$, exchanging $u(a',\omega)$ and $u(a,\omega)$ improves the objective; if $v(a,\omega) = v(a',\omega)$ and $u(a',\omega) > u(a,\omega)$, exchanging leaves the objective unchanged. Thus, we can restrict attention to $0 = u(a_1,\omega) \leq \ldots \leq u(a_n,\omega)$.

For the upper bound, suppose for some $m \in \{2,\ldots,n\}$ that $u(a_m,\omega) > u(a_{m-1},\omega) + \frac{v(a_n,\omega)}{\inf \beta}$. Then for all $i \geq m$, $k \leq m-1$:

$$u(a_i,\omega) \geq u(a_m,\omega) > u(a_{m-1},\omega) + \frac{v(a_n,\omega)}{\inf \beta} \geq u(a_k,\omega) + \frac{v(a_n,\omega)}{\inf \beta}. \tag{17}$$

Denote $W_k := v(a_k,\omega) - \beta u(a_k,\omega)$. Using (14)–(16), the derivative of decreasing $u(a_m,\omega), \ldots, u(a_n,\omega)$ uniformly is

$$\begin{aligned}
-\sum_{i=m}^{n} \frac{dG(u)}{du(a_i,\omega)} &= \sum_{i=m}^{n} \int \frac{\beta}{\lambda} p_\beta(a_i|\omega) \left(\sum_{k=1,k\neq i}^{n} p_\beta(a_k|\omega)\Big[W_k - W_i\Big] \right) dF(\beta) \\
&= \sum_{i=m}^{n} \int \frac{\beta}{\lambda} p_\beta(a_i|\omega) \left(\sum_{k=1}^{m-1} p_\beta(a_k|\omega)\Big[W_k - W_i\Big] \right) dF(\beta) \\
&\geq \sum_{i=m}^{n} \int \frac{\beta}{\lambda} p_\beta(a_i|\omega) \left(\sum_{k=1}^{m-1} p_\beta(a_k|\omega)\Big[\beta\big(u(a_i,\omega) - u(a_k,\omega)\big) - v(a_n,\omega)\Big] \right) dF(\beta),
\end{aligned}$$

which is strictly positive by (17). Hence $u(a_i,\omega) \leq (i-1)\frac{v(a_n,\omega)}{\inf \beta}$, which defines a compact set. □

Lemma 4. *There exists a symmetric utility function $u\colon A \times \Omega \to \mathbb{R}$ that maximizes (12) state-by-state.*

Proof. The objective (12) is symmetric under permutation of actions: permuting both $v(\cdot, \omega)$ and $u(\cdot, \omega)$ by the same permutation leaves the objective unchanged. Thus, if $u(\cdot, \omega)$ solves (12) given $v(\cdot, \omega)$, then permuting $v(\cdot, \omega)$ leads to the same permutation of the solution candidates. Define an equivalence relation on Ω: two states are equivalent if their utility-vectors $v(\cdot, \omega)$ are a permutation of each other. For each equivalence class, pick a representative state and a solution $u(\cdot, \omega)$. For other states ω' in the equivalence class, assign the analogous permutation of $u(\cdot, \omega)$, which is a solution given $v(\cdot, \omega')$. This procedure is well-defined because by Assumption 4 and Lemma 2, all components of $v(\cdot, \omega)$ and $u(\cdot, \omega)$ are pairwise different, so any permutation is uniquely determined. By symmetry of v, the resulting u is symmetric. $\square$

We have thus shown that the inner maximization problem separates into statewise maximization of (12). Next, we show that the overall principal's problem separates by state.

Outer Minimization Problem. Because $p_\beta(a|u(\omega))$ depends only on the statewise utility vector $u(\omega)$ by (9), the outer objective (10) separates by state. Moreover, if a solution to the principal's problem exists, then by symmetry of v and an argument analogous to Lemma 4—using that (10) is symmetric under permutation of actions—a symmetric solution exists. We can therefore drop the symmetry constraint, and the principal's problem separates into statewise problems. It follows that any solution must solve for every state $\omega \in \Omega$ the *statewise principal's problem* (18):

$$\begin{aligned} \min_{u(\cdot,\omega)\colon A \to \mathbb{R}} \int \left(\sum_{a \in A} p_\beta(a|u(\omega)) \phi\big(u(a,\omega) - v(a,\omega)\big) \right) dF(\beta) \\ \text{s.t.} \quad u(\cdot,\omega) \in \operatorname*{arg\,max}_{u(\cdot,\omega)\colon A \to \mathbb{R}} G(u(\omega)). \end{aligned} \tag{18}$$

Lemma 5. *A solution to the statewise principal's problem (18) exists. The solution is unique if the solution to the inner maximization problem is unique up to a common additive constant.*

Proof. Reparametrize $(u(a_1, \omega), \ldots, u(a_n, \omega))$ by $(u_1, \Delta u_2, \ldots, \Delta u_n)$ where $\Delta u_i = u(a_i, \omega) - u(a_1, \omega)$ and $u_1 = u(a_1, \omega)$. The choice probabilities $p_\beta(a_i|u(\omega))$ depend only on $(\Delta u_2, \ldots, \Delta u_n)$, so the inner maximization problem (11) constrains only the Δu_i's, leaving u_1 unrestricted. The outer objective becomes

$$\sum_{i=1}^{n} \phi\big(u_1 + \Delta u_i - v(a_i, \omega)\big) \int p_\beta(a_i|u(\omega)) dF(\beta).$$

By assumption, ϕ is strictly convex and minimized at 0. Thus, for fixed $(\Delta u_2, \ldots, \Delta u_n)$, the objective is strictly convex and radially unbounded in u_1. Since Δu_i is bounded by Lemma 3, this implies a bound on u_1. By Berge's maximum theorem, the set of maximizers from the inner problem is compact. Hence the minimum is attained by continuity.

For uniqueness: if $(\Delta u_2, \ldots, \Delta u_n)$ is unique from the inner problem, then u_1 is unique by strict convexity. $\square$

For $c \in \mathbb{R}$, let $\mathbf{c} := (c, \ldots, c) \in \mathbb{R}^A$.

Lemma 6. *$u(\cdot, \omega) \in \mathbb{R}^A$ solves the statewise principal's problem (18) given $v(\cdot, \omega) \in \mathbb{R}^A$ if and only if $u(\cdot, \omega) + \mathbf{c}$ does so given $v(\cdot, \omega) + \mathbf{c}$.*

Proof. The constraint set (argmax of the inner problem) is unchanged by shifting $v(\cdot, \omega)$ by $\mathbf{c}$, since this shifts the principal's utility by c independent of the agent's choice. The constraint set is also closed under shifting $u(\cdot, \omega)$ by $\mathbf{c}$, since this does not affect choice probabilities. Thus, if $u(\cdot, \omega)$ is in the constraint set given $v(\cdot, \omega)$, then $u(\cdot, \omega) + \mathbf{c}$ is in the constraint set given $v(\cdot, \omega) + \mathbf{c}$.

The outer objective is unchanged if $\mathbf{c}$ is added to both $u(\cdot, \omega)$ and $v(\cdot, \omega)$, since choice probabilities and $\phi(u(a_i, \omega) - v(a_i, \omega))$ are unaffected. Combined with the above, if $u(\cdot, \omega)$ minimizes the socialization cost given $v(\cdot, \omega)$, then $u(\cdot, \omega) + \mathbf{c}$ minimizes it given $v(\cdot, \omega) + \mathbf{c}$. The converse follows from shifting by $-\mathbf{c}$. $\square$

Lemma 7. *There exists a solution u to the principal's problem (10) and a function R, such that for all $(a, \omega) \in A \times \Omega$,*

$$u(a, \omega) = v(a, \omega) + R(\{v(a, \omega) - v(a', \omega)\}_{a' \in A \setminus \{a\}}).$$

Proof. Define a partition $\mathbb{P}$ on the state space Ω based on the equivalence relation that the utility vectors $v(\cdot, \omega)$ of two states are the same up to permutations, shifts of all components by a constant, or combinations of both. For each partition element $P \in \mathbb{P}$, pick a representative state $\omega \in P$ and define $u(\cdot, \omega)$ to be a solution to the statewise principal's problem, which exists by Lemma 5.

For each other state $\omega' \in P$, there exist a unique $c \in \mathbb{R}$ and a unique permutation $\tilde{v}(\cdot, \omega)$ of $v(\cdot, \omega)$ such that $v(\cdot, \omega') = \mathbf{c} + \tilde{v}$ (uniqueness follows from Assumption 4). Define $u(\cdot, \omega')$ to be the analogous permutation of $u(\cdot, \omega)$ plus $\mathbf{c}$. This gives a solution to the statewise principal's problem under ω' by symmetry and Lemma 6. By symmetry of v, the constructed u is symmetric and solves the principal's problem.

By construction, $u(\cdot, \omega)$ is a function only of the material utility vector $v(\cdot, \omega)$. Denote this function by $M \colon \mathbb{R}^A \to \mathbb{R}^n$ and by M_i its i-th component. By construction, permutations of the arguments of M permute the image analogously, that is, M commutes with permutations. Define

$$R(\{v_1, \ldots, v_{n-1}\}) := M_n(-v_1, \ldots, -v_{n-1}, 0),$$

which is well-defined because M_n is invariant under permutations of v_1 through v_{n-1}. By Lemma

6 and M commuting with permutations:

$$\begin{aligned}
u(a_i, \omega) &= M_i(v(a_1, \omega), \ldots, v(a_n, \omega)) \\
&= v(a_i, \omega) + M_i(v(a_1, \omega) - v(a_i, \omega), \ldots, 0, \ldots, v(a_n, \omega) - v(a_i, \omega)) \\
&= v(a_i, \omega) + M_n(v(a_1, \omega) - v(a_i, \omega), \ldots, v(a_n, \omega) - v(a_i, \omega), 0) \\
&= v(a_i, \omega) + R(\{v(a_i, \omega) - v(a', \omega)\}_{a' \in A \setminus \{a_i\}}). \qquad \square
\end{aligned}$$

Note that the constructed solution depends only on the statewise material utilities as well as the distribution over β and λ. Thus, the R is independent of the choice set A. $\square$

C. Theorem 1

Proof. By Theorem 2, there exists a solution with

$$u(a, \omega) = v(a, \omega) + R(v(a, \omega) - v(a_-, \omega)), \tag{19}$$

provided $v(a, \omega) \neq v(a_-, \omega)$ for all states. We first extend this to states where $v(a_1, \omega) = v(a_2, \omega)$, then establish uniqueness of Q and R, and finally verify Properties 1–2 and continuity.

Extension to $v(a_1, \omega) = v(a_2, \omega)$. If $v(a_1, \omega) = v(a_2, \omega)$, the expected material utility in the statewise principal's problem (18) is unaffected by choice probabilities. Entropy is maximized when $p_\beta(a_1|\omega) = p_\beta(a_2|\omega) = 1/2$, achieved iff $u(a_1, \omega) = u(a_2, \omega)$. Since the socialization cost is minimized at 0, the unique solution is $u(a_1, \omega) = u(a_2, \omega) = v(a_1, \omega) = v(a_2, \omega)$. This is consistent with (19) and $R(0) = 0$, and $u = v$ maintains symmetry.

First-order condition. Fix a state $\omega \in \Omega$ and define

$$\begin{aligned}
\Delta u &:= u(a_2, \omega) - u(a_1, \omega) \\
\Delta v &:= v(a_2, \omega) - v(a_1, \omega) \\
\rho_\beta(x) &:= \frac{e^{\frac{\beta}{\lambda} x}}{1 + e^{\frac{\beta}{\lambda} x}}.
\end{aligned}$$

We have $p_\beta(a_2|\omega) = \rho_\beta(\Delta u)$. Using $p_\beta(a_1|\omega) = 1 - p_\beta(a_2|\omega)$, the first-order condition (14) becomes

$$\int \rho'_\beta(\Delta u)\big(\Delta v - \beta \Delta u\big)\, dF(\beta) = 0, \tag{20}$$

which can be rewritten as

$$\Delta u = \Delta v \left(\frac{\int \rho'_\beta(\Delta u) \beta \, dF(\beta)}{\int \rho'_\beta(\Delta u) \, dF(\beta)} \right)^{-1}. \tag{21}$$

The term in parentheses is a weighted average of β, with weights proportional to $\rho'_\beta(\Delta u)$. The logistic choice probabilities allow us to interpret this weighted average as a conditional expectation, which we use to prove uniqueness and starshapedness. This approach exploits the connection between rational inattention and random utility models: under mutual information costs, choice probabilities take the logit form, which is equivalent to a random utility model with extreme value errors. More generally, Csiszár information costs—a broader class that includes mutual information—are equivalent to additive perturbed utility models (Bloedel, Denti and Pomatto 2025), which are in turn equivalent to random utility models in binary decision problems (Fudenberg, Iijima and Strzalecki 2015). The characterization in equation (21) extends to general Csiszár costs, with ρ'_β replaced by the density of the difference of error terms in the associated random utility model.

Uniqueness. Define

$$C(\Delta u) := \frac{\int \rho'_\beta(\Delta u)\beta\Delta u \, dF(\beta)}{\int \rho'_\beta(\Delta u)\, dF(\beta)},$$

so (21) becomes $\Delta v = C(\Delta u)$. Since $\rho_\beta(x)$ is the CDF of Logistic$(0, \lambda/\beta)$, we have $\rho'_\beta(x) = g_{\lambda/\beta}(x)$, where g_s denotes the density of Logistic$(0, s)$. Define independent random variables $\tilde{v} \sim$ Logistic$(0, \lambda)$ truncated to $(0, \infty)$ and $\tilde{\beta} \sim F$, and let $\tilde{u} := \tilde{v}/\tilde{\beta}$. Since $g_{\lambda/\beta}(\Delta u)$ is proportional to the conditional density of $\tilde{u}$ given $\tilde{\beta} = \beta$, the ratio defining C is a posterior mean:

$$C(\Delta u) = \mathbb{E}[\tilde{\beta} \cdot \Delta u \mid \tilde{u} = \Delta u] = \mathbb{E}[\tilde{v} \mid \tilde{u} = \Delta u]$$

for all $\Delta u > 0$. Since $\log \tilde{u} = \log \tilde{v} - \log \tilde{\beta}$, $\log \tilde{u}$ is a location experiment of $\log \tilde{v}$ with noise $-\log \tilde{\beta}$. Since $\log \tilde{\beta}$ is strictly log-concave by assumption, the experiment satisfies the strict monotone likelihood ratio property. Hence the posterior over $\log \tilde{v}$ given $\log \tilde{u} = \log \Delta u$ is strictly increasing in first-order stochastic dominance in $\log \Delta u$, and so the posterior mean $C(\Delta u)$ is strictly increasing on $(0, \infty)$. Since the logistic density is symmetric, C is antisymmetric ($C(-\Delta u) = -C(\Delta u)$). With $C(\Delta u) > 0$ for $\Delta u > 0$, this implies that C is strictly increasing on $\mathbb{R}$. Thus Δv is uniquely determined by Δu. Since Q maps $\Delta v \mapsto \Delta u$, Q is unique, and by Lemma 5, so is R. Thus (21) characterizes the solution Q.[18]

Property 2: Starshapedness. By (21), $\Delta u = 0$ when $\Delta v = 0$, so $Q(0) = 0$. Moreover, since the term in parentheses in (21) is a weighted average of $\beta > 0$, we have $\Delta u > 0$ when $\Delta v > 0$, so $Q(\Delta v) > 0$ for $\Delta v > 0$. By Definition 2, it remains to show $Q(k\Delta v) > kQ(\Delta v)$ for $\Delta v > 0$ and $k > 1$. This is equivalent to $Q(\Delta v)/\Delta v$ being strictly increasing on $\mathbb{R}_{>0}$, or equivalently, $Q^{-1}(\Delta u)/\Delta u$ being strictly decreasing on $\mathbb{R}_{>0}$ (using that Q is strictly increasing by the uniqueness

[18] Even without the assumption that $\log \beta$ is strictly log-concave, one obtains a weaker form of uniqueness. Under two actions, the objective (13) is strictly supermodular in Δu and Δv, so the solutions Δu are strictly increasing in the strong set order. Thus, Q is non-decreasing and wherever non-unique, it has an upward jump. Since a non-decreasing function can have only countably many jump discontinuities, Q and hence R are unique almost everywhere on $\mathbb{R}$.

proof). By (21), $Q^{-1} = C$, so

$$\frac{Q^{-1}(\Delta u)}{\Delta u} = \frac{C(\Delta u)}{\Delta u} = \mathbb{E}[\tilde{\beta} \mid \tilde{u} = \Delta u].$$

Since $\log \tilde{u} = \log \tilde{v} - \log \tilde{\beta}$, $\log \tilde{u}$ is a location experiment of $-\log \tilde{\beta}$ with noise $\log \tilde{v}$. The density of the logistic distribution is strictly log-concave and decreasing on $(0, \infty)$, which implies that $\log \tilde{v}$ has a strictly log-concave density (An 1997, Proposition 4; the strict version follows by the same proof). By the strict monotone likelihood ratio property, the posterior over $-\log \tilde{\beta}$ given $\log \tilde{u} = \log \Delta u$ is strictly increasing in first-order stochastic dominance in $\log \Delta u$. Since $\tilde{\beta}$ is decreasing in $-\log \tilde{\beta}$, the posterior over $\tilde{\beta}$ is strictly decreasing in first-order stochastic dominance, hence $\mathbb{E}[\tilde{\beta} \mid \tilde{u} = \Delta u]$ is strictly decreasing in Δu.

Property 1: $R(x) \gtreqless 0 \Leftrightarrow x \gtreqless 0$**.** Given $Q(x)$, the values $R(x)$ and $R(-x)$ minimize the socialization cost subject to $x + R(x) - R(-x) = Q(x)$:

$$\begin{aligned} \min_{R(x), R(-x) \in \mathbb{R}} \left(\int \rho_\beta(Q(x))\, dF(\beta) \right) \phi(R(x)) + \left(1 - \int \rho_\beta(Q(x))\, dF(\beta)\right) \phi(R(-x)) \qquad (22) \\ \text{s.t.} \quad x + R(x) - R(-x) = Q(x). \end{aligned}$$

We have $R(0) = 0$ from above. For $x > 0$, equation (21) implies $Q(x) - x > 0$ by $\beta \leq 1$.[19] If $R(x) \leq 0$, then $R(-x) = R(x) - (Q(x) - x) < 0$. Since ϕ is strictly convex with minimum at 0 and $\phi'(0) = 0$, increasing both $R(x)$ and $R(-x)$ by a small $\varepsilon > 0$ preserves the constraint and decreases the objective. Hence $R(x) > 0$. The case $x < 0$ is analogous.

Continuity. By Lemma 3, we can restrict Δu to lie in a compact set that depends continuously on Δv. Since the objective (12) is jointly continuous in $(\Delta u, \Delta v)$, Berge's maximum theorem and uniqueness imply Q is continuous.

For R: using the constraint, rewrite (22) as minimizing over $R(x)$ alone:

$$\min_{R(x) \in \mathbb{R}} \left(\int \rho_\beta(Q(x))\, dF(\beta) \right) \phi(R(x)) + \left(1 - \int \rho_\beta(Q(x))\, dF(\beta)\right) \phi(Q(x) - x - R(x)).$$

By Property 1, we can restrict attention to $R(x) \in [0, Q(x) - x]$ for $x \geq 0$ and $R(x) \in [Q(x) - x, 0]$ for $x \leq 0$. This compact-valued correspondence is continuous in x since Q is continuous. The objective is jointly continuous in $(x, R(x))$ since Q, ρ_β, and ϕ are continuous. By Berge's maximum theorem and uniqueness, R is continuous. $\square$

[19] This is the only place in the paper where we use $\beta \leq 1$. The implication $Q(x) > x$ and hence Property 1 also follows from the weaker condition $\mathbb{E}[\beta^2] \leq \mathbb{E}[\beta]$. This can be seen as follows. Since $Q(x)/x$ is increasing by starshapedness, it is sufficient to show that $Q(x)/x$ converges to a number greater or equal to 1 as $x \to 0$. By Remark 1 and the first part of Proposition 3, $Q(x)/x \to \mathbb{E}[\beta^2]/\mathbb{E}[\beta] \cdot x$, which is greater or equal to 1 if and only if $\mathbb{E}[\beta^2] \leq \mathbb{E}[\beta]$.

D. Proposition 1

Proof. Define $\Psi(x,y) := Q(\bar{v}(x) - \bar{v}(y))$. Since Q is skew-symmetric and continuous and $\bar{v}$ is continuous, Ψ is skew-symmetric and continuous, so preferences have a generalized regret representation. Starshapedness and skew-symmetry together imply that Q is strictly increasing. Since $\bar{v}$ and Q are strictly increasing, Ψ is strictly increasing in its first argument, establishing increasingness. For convexity, let $x > y > z$ and define $a := \bar{v}(x) - \bar{v}(y) > 0$ and $b := \bar{v}(y) - \bar{v}(z) > 0$. Then $\Psi(x,z) = Q(a+b)$ and $\Psi(x,y) + \Psi(y,z) = Q(a) + Q(b)$. For nonnegative functions on $[0,\infty)$ that vanish at 0, starshapedness implies superadditivity (Bruckner and Ostrow 1962), so $Q(a+b) > Q(a) + Q(b)$.[20] $\square$

E. Proposition 2

Proof. Loomes and Sugden (1982) show that the Allais paradox and the common-ratio effect are equivalent to strict superadditivity of Q,

$$\forall x_1, x_2 \colon 0 < x_2 < x_1 \to Q(x_1) - Q(x_1 - x_2) - Q(x_2) > 0, \tag{23}$$

and that simultaneous gambling and insurance is equivalent to

$$\forall x > 0 \colon Q\left(\tfrac{p}{1-p}x\right) \gtreqless \tfrac{p}{1-p}Q(x) \quad \text{if } p \gtreqless 0.5. \tag{24}$$

As established above, our definition of starshapedness implies strict superadditivity, hence (23). Condition (24) is equivalent to starshapedness: substituting $k = \frac{p}{1-p}$, we have $k > 1$ if and only if $p > 0.5$, and (24) becomes $Q(kx) > kQ(x)$ for $k > 1$. $\square$

F. Choice under Equilibrium Beliefs

It is useful to recall the distinction between the attention stage (choice of a signal structure) and the decision-making stage (choice of an action conditional on the signal). A common theme in the principal-expert literature is that the principal might want to distort the expert's incentives at the decision-making stage to incentivize more information acquisition, as already pointed out by Demski and Sappington (1987). The following result shows that in our model, this is not necessary: under equilibrium beliefs, the agent's choice is undistorted.

Observation 1. *Under the beliefs that agents acquire in equilibrium, the agent's choice coincides with the principal's preferred action at that belief.*

Proof. Fix any type β. We show that the posterior $\gamma_\beta^{a_1}$ this type holds when choosing a_1 makes a_1 optimal for the principal. Since the principal's payoff from action a under state ω is $v(a,\omega)$, we need to show

$$\mathbb{E}_{\gamma_\beta^{a_1}}[\Delta v] := \sum_{\omega \in \Omega} \gamma_\beta^{a_1}(\omega) \Delta v(\omega) > 0$$

[20] Bruckner and Ostrow (1962) prove this for weak inequalities; the strict version follows by the same argument.

where $\Delta v(\omega) = v(a_1, \omega) - v(a_2, \omega)$.

By symmetry, $p_\beta(a_1) = 1/2$, so the posterior is

$$\gamma_\beta^{a_1}(\omega) = \frac{\mu(\omega)\, p_\beta(a_1|\omega)}{p_\beta(a_1)} = 2\mu(\omega)\, p_\beta(a_1|\omega).$$

For $x \in \mathbb{R}$, let $\Omega_x = \{\omega \in \Omega : \Delta v(\omega) = x\}$ denote the set of states with material utility difference x. By symmetry, $\mu(\Omega_x) = \mu(\Omega_{-x})$ for all x. We pair such sets and show that each pair contributes positively to $\mathbb{E}_{\gamma_{a_1}^\beta}[\Delta v]$.

Let $x > 0$. Since Q is strictly increasing with $Q(0) = 0$, we have $Q(x) > 0 > Q(-x)$, so for all $\omega \in \Omega_x, \omega' \in \Omega_{-x}$

$$p_\beta(a_1|\omega) = p_\beta(Q(x)) > \tfrac{1}{2} > p_\beta(-Q(x)) = p_\beta(a_1|\omega')$$

where, with slight abuse of notation, $p_\beta(x) = e^{\frac{\beta}{\lambda}x}/(1 + e^{\frac{\beta}{\lambda}x})$. The contribution of Ω_x and Ω_{-x} to $\mathbb{E}_{\gamma_\beta^{a_1}}[\Delta v]$ is

$$\sum_{\omega \in \Omega_x} \gamma_\beta^{a_1}(\omega) \Delta v(\omega) + \sum_{\omega' \in \Omega_{-x}} \gamma_\beta^{a_1}(\omega') \Delta v(\omega') = 2\mu(\Omega_x)\big(p_\beta(Q(x)) - p_\beta(-Q(x))\big) \cdot x > 0.$$

Summing over all $x > 0$, we obtain $\mathbb{E}_{\gamma_\beta^{a_1}}[\Delta v] > 0$. The argument for a_2 is analogous. □

G. Proposition 3

We first establish a scaling property for the statewise maximization problem, which applies to any number of actions.

Lemma 8 (Scaling)**.** *Consider the statewise maximization problem* (8) *with material utilities $v(\cdot, \omega)$ and complexity λ. If $u(\cdot, \omega)$ is a solution, then $k \cdot u(\cdot, \omega)$ is a solution to the problem with material utilities $k \cdot v(\cdot, \omega)$ and complexity $k\lambda$, for any $k > 0$.*

Proof. The choice probabilities depend on (u, λ) only through the ratio $\beta u/\lambda$:

$$p_\beta(a|u(\omega)) = \frac{e^{\frac{\beta}{\lambda}u(a,\omega)}}{\sum_{a' \in A} e^{\frac{\beta}{\lambda}u(a',\omega)}}.$$

Under complexity $k\lambda$, the choice probabilities at ku satisfy $p_\beta(a|ku(\omega); k\lambda) = p_\beta(a|u(\omega); \lambda)$, so the choice probabilities are unchanged. The objective (8) under $(kv, k\lambda)$ evaluated at ku is:

$$\int \left(k\lambda\, H(p_\beta(ku)) + \sum_{a \in A} p_\beta(a|ku) \cdot kv(a, \omega) \right) dF(\beta) = k \int \left(\lambda\, H(p_\beta(u)) + \sum_{a \in A} p_\beta(a|u) \cdot v(a, \omega) \right) dF(\beta).$$

Since scaling the objective by a positive constant preserves maximizers, ku solves the problem under $(kv, k\lambda)$ if and only if u solves it under (v, λ). □

Proof of Proposition 3. (i) By Lemma 8, scaling (v, λ) to $(kv, k\lambda)$ scales the statewise solution from u to ku. The principal's objective and the agent's incentive constraint both scale by k, preserving maximizers. The symmetry constraint is also preserved since scaling all material utilities by the same constant maintains symmetry. Therefore, if u solves (P) under (v, λ), then ku solves (P) under $(kv, k\lambda)$.

Under two actions, Q_λ maps Δv to Δu, so $Q_{k\lambda}(k\Delta v) = k \cdot Q_\lambda(\Delta v)$. Setting $k = \lambda'/\lambda$ and $x = k\Delta v$ yields the scaling property.

(ii) The comparative static on Q follows from (i) and starshapedness. For $\lambda' < \lambda$, let $k = \lambda'/\lambda < 1$. By the scaling property:

$$Q_{\lambda'}(x) = k\, Q_\lambda(x/k).$$

Since $1/k > 1$ and $x > 0$, starshapedness of Q_λ implies $Q_\lambda(x/k)/(x/k) > Q_\lambda(x)/x$, so $Q_\lambda(x/k) > (1/k)\, Q_\lambda(x)$. Therefore:

$$Q_{\lambda'}(x) = k\, Q_\lambda(x/k) > k \cdot \frac{1}{k}\, Q_\lambda(x) = Q_\lambda(x).$$

For the comparative static on $|R(-x)|$, we use supermodularity. Recall from the proof of Theorem 1 that $R(x)$ and $R(-x)$ minimize the expected socialization cost subject to being consistent with $Q(x)$:

$$\begin{aligned} \min_{R(x), R(-x) \in \mathbb{R}} \quad & q \cdot \phi(R(x)) + (1-q) \cdot \phi(R(-x)) \\ \text{s.t.} \quad & R(x) - R(-x) = Q(x) - x, \end{aligned}$$

where $q := \int p_\beta(Q(x))\, dF(\beta)$ is the probability of taking the right action and $p_\beta(y) = e^{\frac{\beta}{\lambda} y}/(1+e^{\frac{\beta}{\lambda} y})$. Substituting the constraint $R(x) = Q(x) - x + R(-x)$, the problem reduces to minimizing over $R(-x)$:

$$\Phi(R(-x); q, Q(x)) := q \cdot \phi(Q(x) - x + R(-x)) + (1-q) \cdot \phi(R(-x)).$$

When λ decreases to λ', both $Q(x)$ increases (as just shown) and q increases, since p_β is strictly increasing in its argument and q depends on $Q(x)/\lambda$. We show that Φ has strictly increasing differences in $(R(-x), q)$ and in $(R(-x), Q(x))$. Recall from Theorem 1 that at the optimum, $R(-x) < 0$ and $R(x) = Q(x) - x + R(-x) > 0$.

For the cross-partial in $(R(-x), q)$:

$$\frac{\partial^2 \Phi}{\partial R(-x)\, \partial q} = \phi'(Q(x) - x + R(-x)) - \phi'(R(-x)).$$

Since ϕ is strictly convex and minimized at 0, we have $\phi'(y) > 0$ for $y > 0$ and $\phi'(y) < 0$ for $y < 0$. Thus $\phi'(Q(x) - x + R(-x)) > 0$ and $\phi'(R(-x)) < 0$, so the cross-partial is strictly positive.

For the cross-partial in $(R(-x), Q(x))$:

$$\frac{\partial^2 \Phi}{\partial R(-x)\, \partial Q(x)} = q \cdot \phi''(Q(x) - x + R(-x)) > 0,$$

since $q > 0$ and $\phi'' > 0$ by strict convexity.

By Topkis' theorem, the optimal $R(-x)$ is strictly decreasing in both q and $Q(x)$. Since $R(-x) < 0$, this means $|R(-x)|$ is strictly increasing in both parameters, and hence strictly increasing when λ decreases. □

H. Alternative Model Interpretations

H.1 Biological Evolution

The main text interprets the model through the lens of parental socialization. Here we sketch an alternative interpretation based on biological evolution. The model follows the principal-agent approach to the evolution of preferences (Robson 2001b; Samuelson and Swinkels 2006; Rayo and Becker 2007). This approach likens evolution to a principal choosing preferences to incentivize fitness-maximizing behavior. The principal's maximization stands for the outcome of an evolutionary process where preferences are heritable and preferences that lead to higher fitness dominate the population over time.

The bias toward insufficient attention can be understood again as present bias. Present bias is deeply rooted: both humans and non-human animals exhibit it (Ainslie 1992), suggesting that time preferences are difficult to change through evolution. Emotions like regret may thus have evolved as a second-best response: rather than eliminating present bias directly, evolution shaped emotional responses to decision outcomes that partially compensate for it.

This view of emotions as incremental fixes is consistent with the patchwork nature of evolution (Jacob 1977) and accounts of the brain as a "kludge" assembled from older and newer components (Linden 2007; Marcus 2008). Moreover, once regret is in place, removing the bias would lead to excessive attention. Evolution can get stuck in an equilibrium where regret responds optimally to the bias and the bias responds optimally to regret. Waldman (1994) formally shows that such a second-best equilibrium can be evolutionarily stable, even if both traits—regret and present bias—can evolve at the same time. The reason is that even if a mutant possessed no regret nor bias, their offspring would have suboptimal trait combinations due to genetic recombination with the incumbent population's traits.

The two key constraints in the model—that the agent's bias and attention are not directly observable—also have natural interpretations under the biological account. Actions and outcomes are easier to recall and verify than pre-decision variables like temptation intensity or attention paid. This asymmetry is central to self-signaling models in economics (Bodner and Prelec 2003; Bénabou and Tirole 2004). Such within-individual information asymmetries are not uncommon in psychology (e.g., research on self-deception) and neuroscience, where brain modularity and limited neural connectivity imply that not all information is globally accessible (Bodner and Prelec 2003; Brocas and Carrillo 2008).

H.2 Contracting with an Expert

The model can be interpreted as a principal hiring an agent (expert) to acquire information and make decisions on the principal's behalf, for instance, in CEO or fund manager compensation. The principal reaps utility $v(a, \omega)$ from action a under state ω and pays monetary transfers $m(a, \omega)$ to the agent, valued quasi-linearly by both. The agent's bias β can capture either a conflict of interest or a behavioral bias. We discuss both cases in turn below and show that despite apparent differences in what principal and agent care about, the models are equivalent to versions of our main model.

In both cases, our results contribute to understanding compensation schemes that evaluate the agent's performance relative to a peer benchmark, which is common for fund managers (Evans, Gómez, Ma, and Tang 2024) and, to some degree, CEOs (Edmans, Gabaix, and Jenter 2017). Such benchmarks can be justified by Holmström's (1979) informativeness criterion in hidden-action settings, because subtracting exogenous shocks to the industry increases the informativeness of the performance evaluation. Our model offers an alternative explanation based on incentives for information acquisition: peer comparisons can be interpreted as counterfactual comparisons, since material utilities of relevant peers inform the principal about the payoff that alternative actions would have led to.

Agent with Conflicting Interests Assume that both principal and agent care about information costs and decision outcomes but weigh them differently, as in Szalay (2005). For instance, investors may internalize the time cost of information gathering due to decision delay or diversion from other activities, and CEOs or fund managers care about decision outcomes due to intrinsic motivation or reputation. Still, from the principal's perspective, the agent may overweight information costs absent financial incentives. We assume the principal is uncertain about how much the agent overweights information costs. Information acquisition is unverifiable but the decision outcome (a, ω) is contractible.

Furthermore, we assume the principal faces an ex-ante budget constraint (BC) requiring that expected transfers equal some $b \in \mathbb{R}$. To resolve indifferences, we assume the principal minimizes the variance of transfers, for example due to a secondary concern for transaction costs. As in the main model, this refinement pins down statewise constants to transfers, but is otherwise immaterial. Formally, the principal chooses the solution with minimal variance of transfers to the following problem:

$$\max_{m,(p_\beta)_\beta} \int \left(\sum_{\omega,a} \mu(\omega) p_\beta(a|\omega) \big(v(a,\omega) - m(a,\omega) \big) - \lambda c(p_\beta) \right) dF(\beta)$$

$$\text{s.t.}$$

$$\forall \beta: \;\; p_\beta \in \arg\max_p \;\; \beta \sum_{\omega,a} \mu(\omega) p(a|\omega)(v(a,\omega) + m(a,\omega)) - \lambda c(p) \tag{IC}$$

$$\int \sum_{\omega,a} \mu(\omega) p_\beta(a|\omega) m(a,\omega)\, dF(\beta) = b \tag{BC}$$

$$v + m \text{ is symmetric.} \tag{SYM}$$

By (BC), expected transfers equal the constant b, so the $-m(a,\omega)$ term in the principal's objective contributes only the constant $-b$. Thus, the objective and (IC) are equivalent to the main model. Since shifting all transfers by a constant does not affect the agent's behavior, a solution to this problem with budget b is given by taking a solution with budget $b = 0$ and shifting all transfers by b. Thus, this problem is equivalent to the principal's problem (P) under a quadratic socialization cost $\phi(x) = x^2$, up to shifting all transfers by b. To see this, note that $\mathbb{E}[X^2] = \mathbb{E}[X]^2 + \mathrm{Var}[X]$, so minimizing the socialization cost $\Phi(m)$ is equivalent to minimizing the sum of squared expected transfers and variance of transfers. The principal can shift all transfers to achieve $\mathbb{E}[m] = 0$, so the refinement of minimizing the quadratic socialization cost is equivalent to the budget constraint (BC) with $b = 0$ plus the refinement of minimizing the variance of transfers.

Dynamically Inconsistent Agent Consider a profit-maximizing principal who hires a dynamically inconsistent agent.[21] The principal values the decision outcome net of transfers; the agent values transfers net of attention costs. After signing the contract, the agent faces a stochastic temptation shock to his preferences, which takes the form of a present bias. The principal faces a participation constraint (IR) guaranteeing expected utility U_0 to the agent before the bias realizes:

$$\max_{m,(p_\beta)_\beta} \int \left(\sum_{\omega,a} \mu(\omega) p_\beta(a|\omega) \big(v(a,\omega) - m(a,\omega) \big) \right) dF(\beta)$$

$$\text{s.t.}$$

$$\forall \beta: \;\; p_\beta \in \arg\max_p \;\; \beta \sum_{\omega,a} \mu(\omega) p(a|\omega) m(a,\omega) - \lambda c(p) \tag{IC}$$

$$\int \left(\sum_{\omega,a} \mu(\omega) p_\beta(a|\omega) m(a,\omega) - \lambda c(p_\beta) \right) dF(\beta) \geq U_0 \tag{IR}$$

$$m \text{ is symmetric.} \tag{SYM}$$

By standard arguments, (IR) binds. Substituting the binding (IR) into the principal's objective

[21] See Kaur, Kremer, and Mullainathan (2015) for evidence on self-control issues at work.

eliminates transfers, leaving material utility net of attention cost (minus U_0). Identifying m with u from the main text, the objective and (IC) are identical: here optimal transfers emulate v in addition to emotions since the agent has no direct interest in the decision. The principal can add a uniform constant to transfers to satisfy the binding (IR) constraint, so (IR) acts as an ex-ante budget constraint (BC) with b depending on U_0. Adding the refinement that the principal minimizes the variance of transfers renders the problem equivalent to ours under a quadratic socialization cost.

Online Appendix

A. Proposition 4

Proof. We cannot apply upper hemicontinuity from Berge's maximum theorem because our objective is not defined at $\lambda = 0$ or $\lambda = \infty$. Instead, we work directly with the first-order conditions for optimality.

Since the principal's problem separates by state, it suffices to prove convergence for the statewise problem. Fix $\omega \in \Omega$ and adopt the shorthand $u_i := u(a_i, \omega)$ and $v_i := v(a_i, \omega)$ for $i = 1, \ldots, n$. By Assumption 4, assume $v_1 < \cdots < v_n$. By Lemma 2, $u_1 < \cdots < u_n$. Normalize $u_n = v_n = 0$ and define $u := (u_1, ..., u_{n-1})$ and $v := (v_1, ..., v_{n-1})$. Since we vary λ, we make the dependence on λ explicit and write $p^i_{\beta,\lambda}(u)$ for the choice probability $p_\beta(a_i|\omega)$.

The first-order condition for the statewise maximization problem with respect to u_i, derived from equations (14)–(16), is

$$\int \left(\frac{\beta}{\lambda} p^i_{\beta,\lambda}(u)(1 - p^i_{\beta,\lambda}(u))(v_i - \beta u_i) - \sum_{j=1, j\neq i}^{n-1} \frac{\beta}{\lambda} p^i_{\beta,\lambda}(u) p^j_{\beta,\lambda}(u)(v_j - \beta u_j) \right) dF(\beta) = 0. \tag{25}$$

We prove the two limits separately, starting with the simpler $\lambda \to \infty$ case.

Case $\lambda \to \infty$. *Step 1: Limiting system of first-order conditions.* We multiply the first-order conditions (25) by λ and write them in matrix form as $\mathcal{A}(u, \lambda)v = \mathcal{B}(u, \lambda)u$, where $\mathcal{A}$ and $\mathcal{B}$ are $(n-1) \times (n-1)$ matrices with entries

$$\mathcal{A}_{ij}(u, \lambda) = \begin{cases} \int \beta \, p^i_{\beta,\lambda}(u)(1 - p^i_{\beta,\lambda}(u)) \, dF(\beta) & \text{if } i = j \\ -\int \beta \, p^i_{\beta,\lambda}(u) p^j_{\beta,\lambda}(u) \, dF(\beta) & \text{if } i \neq j \end{cases}$$

$$\mathcal{B}_{ij}(u, \lambda) = \begin{cases} \int \beta^2 \, p^i_{\beta,\lambda}(u)(1 - p^i_{\beta,\lambda}(u)) \, dF(\beta) & \text{if } i = j \\ -\int \beta^2 \, p^i_{\beta,\lambda}(u) p^j_{\beta,\lambda}(u) \, dF(\beta) & \text{if } i \neq j \end{cases}$$

As $\lambda \to \infty$, we have $p^i_{\beta,\lambda}(u) \to 1/n$ for all i, β, and u. Thus

$$p^i_{\beta,\lambda}(u)(1 - p^i_{\beta,\lambda}(u)) \to \frac{1}{n} \cdot \frac{n-1}{n} = \frac{n-1}{n^2}, \qquad p^i_{\beta,\lambda}(u) p^j_{\beta,\lambda}(u) \to \frac{1}{n^2}.$$

The matrices converge to

$$\mathcal{A}(u, \lambda) \to \mathbb{E}[\beta] \cdot M,$$
$$\mathcal{B}(u, \lambda) \to \mathbb{E}[\beta^2] \cdot M,$$

where M is the $(n-1) \times (n-1)$ matrix with $(n-1)/n^2$ on the diagonal and $-1/n^2$ off the diagonal.

The limiting system of first-order conditions is

$$\mathbb{E}[\beta] \cdot Mv = \mathbb{E}[\beta^2] \cdot Mu.$$

Since M is invertible (it is strictly diagonally dominant), this simplifies to $v_i = (\mathbb{E}[\beta^2]/\mathbb{E}[\beta])u_i$ for each $i = 1, \ldots, n-1$. The unique solution is $u_i^* := (\mathbb{E}[\beta]/\mathbb{E}[\beta^2])v_i$.

Step 2: Convergence. Reparametrize with $l = 1/\lambda$, so $\lambda \to \infty$ corresponds to $l \to 0$. With slight abuse of notation, write the matrices as functions of l and the system of first-order conditions as $F(u, v, l) := \mathcal{A}(u, l)v - \mathcal{B}(u, l)u = 0$, which is jointly continuous in (u, v, l) including at $l = 0$. At $l = 0$, the limiting system has the unique solution $u^*(v) := (\mathbb{E}[\beta]/\mathbb{E}[\beta^2])v$.

Step 2: Convergence. Reparametrize with $l = 1/\lambda$, so $\lambda \to \infty$ corresponds to $l \to 0$. With slight abuse of notation, write the matrices as functions of l and the system of first-order conditions as $F(u, v, l) := \mathcal{A}(u, l)v - \mathcal{B}(u, l)u = 0$, which is jointly continuous in (u, v, l) including at $l = 0$. At $l = 0$, the limiting system has the unique solution $u^*(v) := (\mathbb{E}[\beta]/\mathbb{E}[\beta^2])v$ by Step 1.

We show that solutions u to the $F(u, v, l) = 0$ converge to $u^*(v)$ as $l \to 0$, uniformly over v in any compact set $V \subseteq \mathbb{R}^{n-1}$. By the proof of Lemma 3, solutions lie in a compact set K_V depending only on V. Joint continuity of F implies that the solution correspondence has closed graph. Together with compactness of K_V, this implies upper hemicontinuity. Since the limiting system has a unique solution $u^*(v)$, which is continuous in v, upper hemicontinuity implies that solutions converge to $u^*(v)$, uniformly over $v \in V$.

Case $\lambda \to 0$. *Step 1: Normalizing the first-order conditions.* Since $u_n = 0 > u_j$ for all $j < n$, $p^n_{\beta,\lambda}(u) \to 1$ and $p^j_{\beta,\lambda}(u) \to 0$ for all $j < n$ as $\lambda \to 0$. Dividing the first-order condition (25) for row i by $\int(\beta/\lambda)p^i_{\beta,\lambda}(u)(1 - p^i_{\beta,\lambda}(u))\, dF(\beta)$, we obtain:

$$\begin{aligned} v_i + \sum_{j \neq i,n} & \frac{\int(\beta/\lambda)p^i_{\beta,\lambda}(u)p^j_{\beta,\lambda}(u)\, dF(\beta)}{\int(\beta/\lambda)p^i_{\beta,\lambda}(u)(1 - p^i_{\beta,\lambda}(u))\, dF(\beta)} v_j \\ &= \frac{\int(\beta^2/\lambda)p^i_{\beta,\lambda}(u)(1 - p^i_{\beta,\lambda}(u))\, dF(\beta)}{\int(\beta/\lambda)p^i_{\beta,\lambda}(u)(1 - p^i_{\beta,\lambda}(u))\, dF(\beta)} u_i + \sum_{j \neq i,n} \frac{\int(\beta^2/\lambda)p^i_{\beta,\lambda}(u)p^j_{\beta,\lambda}(u)\, dF(\beta)}{\int(\beta/\lambda)p^i_{\beta,\lambda}(u)(1 - p^i_{\beta,\lambda}(u))\, dF(\beta)} u_j. \end{aligned}$$

We show: (i) the coefficients on v_j and u_j for $j \neq i, n$ vanish as $\lambda \to 0$, and (ii) the coefficient on u_i converges to $\underline{\beta}$.

Step 2: Off-diagonal coefficients vanish. Consider the coefficient on v_j for $j \neq i, n$:

$$\frac{\int(\beta/\lambda)p^i_{\beta,\lambda}(u)p^j_{\beta,\lambda}(u)\, dF(\beta)}{\int(\beta/\lambda)p^i_{\beta,\lambda}(u)(1 - p^i_{\beta,\lambda}(u))\, dF(\beta)} = \frac{\int \frac{p^j_{\beta,\lambda}(u)}{1 - p^i_{\beta,\lambda}(u)} \cdot (\beta/\lambda)p^i_{\beta,\lambda}(u)(1 - p^i_{\beta,\lambda}(u))\, dF(\beta)}{\int(\beta/\lambda)p^i_{\beta,\lambda}(u)(1 - p^i_{\beta,\lambda}(u))\, dF(\beta)}.$$

This is a weighted average of $p^j_{\beta,\lambda}(u)/(1-p^i_{\beta,\lambda}(u))$. Since $u_j < 0$:

$$p^j_{\beta,\lambda}(u) = \frac{e^{\beta u_j/\lambda}}{\sum_{m=1}^n e^{\beta u_m/\lambda}} \le e^{\beta u_j/\lambda} \le e^{\underline{\beta} u_j/\lambda} \to 0$$

as $\lambda \to 0$, uniformly over $\beta \in \operatorname{supp}(\beta)$. For the denominator:

$$1 - p^i_{\beta,\lambda}(u) \ge p^n_{\beta,\lambda}(u) = \frac{1}{\sum_{m=1}^n e^{\beta u_m/\lambda}} \ge \frac{1}{n},$$

since each term in the sum is at most 1 (as $u_m \le u_n = 0$ for all m). Thus $p^j_{\beta,\lambda}(u)/(1-p^i_{\beta,\lambda}(u)) \le n \cdot e^{\underline{\beta} u_j/\lambda} \to 0$ uniformly over β. Since the coefficient is a weighted average of a quantity converging uniformly to 0, the coefficient vanishes. The same argument applies to the coefficient on u_j.

Step 3: Diagonal coefficient converges to $\underline{\beta}$. The coefficient on u_i is:

$$\frac{\int (\beta^2/\lambda) p^i_{\beta,\lambda}(u)(1-p^i_{\beta,\lambda}(u))\, dF(\beta)}{\int (\beta/\lambda) p^i_{\beta,\lambda}(u)(1-p^i_{\beta,\lambda}(u))\, dF(\beta)} = \frac{\int g_\lambda(\beta)\beta\, dF(\beta)}{\int g_\lambda(\beta)\, dF(\beta)},$$

where $g_\lambda(\beta) := (\beta/\lambda)p^i_{\beta,\lambda}(u)(1-p^i_{\beta,\lambda}(u))$. Fix $\varepsilon > 0$ and decompose:

$$\frac{\int g_\lambda(\beta)\beta\, dF(\beta)}{\int g_\lambda(\beta)\, dF(\beta)} = \frac{A(\lambda)+B(\lambda)}{1+C(\lambda)},$$

where

$$A(\lambda) := \frac{\int_{\underline{\beta}}^{\underline{\beta}+\varepsilon} g_\lambda(\beta)\beta\, dF(\beta)}{\int_{\underline{\beta}}^{\underline{\beta}+\varepsilon} g_\lambda(\beta)\, dF(\beta)}, \quad B(\lambda) := \frac{\int_{\underline{\beta}+\varepsilon}^{1} g_\lambda(\beta)\beta\, dF(\beta)}{\int_{\underline{\beta}}^{\underline{\beta}+\varepsilon} g_\lambda(\beta)\, dF(\beta)},$$
$$C(\lambda) := \frac{\int_{\underline{\beta}+\varepsilon}^{1} g_\lambda(\beta)\, dF(\beta)}{\int_{\underline{\beta}}^{\underline{\beta}+\varepsilon} g_\lambda(\beta)\, dF(\beta)}.$$

We have $\underline{\beta} \le A(\lambda) \le \underline{\beta}+\varepsilon$ because $A(\lambda)$ is a weighted average of β over $[\underline{\beta}, \underline{\beta}+\varepsilon]$. To show $B(\lambda) \to 0$ and $C(\lambda) \to 0$, we establish an exponential bound on $g_\lambda(\beta)$. For $\beta > \underline{\beta}+\varepsilon$:

$$\frac{p^i_{\beta,\lambda}(u)}{p^i_{\underline{\beta}+\varepsilon,\lambda}(u)} = e^{(\beta-(\underline{\beta}+\varepsilon))u_i/\lambda} \cdot \frac{\sum_{m=1}^n e^{(\underline{\beta}+\varepsilon)u_m/\lambda}}{\sum_{m=1}^n e^{\beta u_m/\lambda}} \le n \cdot e^{(\beta-(\underline{\beta}+\varepsilon))u_i/\lambda},$$

where the first factor decays exponentially in β since $u_i < 0$, and the second factor is bounded by n. Combined with $(1-p^i_{\beta,\lambda}(u))/(1-p^i_{\underline{\beta}+\varepsilon,\lambda}(u)) \le n$ and $\beta/(\underline{\beta}+\varepsilon) \le 1/\underline{\beta}$:

$$g_\lambda(\beta) \le \frac{n^2}{\underline{\beta}} \cdot g_\lambda(\underline{\beta}+\varepsilon) \cdot e^{(\beta-\underline{\beta}-\varepsilon)u_i/\lambda}.$$

Thus:

$$B(\lambda) \leq \frac{n^2/\underline{\beta}}{F(\underline{\beta}+\varepsilon)} \int_{\underline{\beta}+\varepsilon}^{1} e^{(\beta-\underline{\beta}-\varepsilon)u_i/\lambda} \beta \, dF(\beta).$$

The integrand converges pointwise to 0 for all $\beta > \underline{\beta} + \varepsilon$ and $u < 0$, and is bounded by 1. By dominated convergence, $B(\lambda) \to 0$. Similarly, $C(\lambda) \to 0$.

Thus $\limsup_{\lambda \to 0} \frac{\int g_\lambda(\beta)\beta \, dF(\beta)}{\int g_\lambda(\beta) \, dF(\beta)} \leq \underline{\beta} + \varepsilon$. Since $\varepsilon > 0$ was arbitrary and the ratio is bounded below by $\underline{\beta}$, the limit equals $\underline{\beta}$.

Step 4: Solutions remain bounded away from 0*.* The convergence arguments in Steps 2–3 require u_i to be bounded away from 0 for the exponential bounds to apply uniformly. We now establish this.

Suppose, for contradiction, that along some sequence $\lambda_k \to 0$, we have solutions to the first-order conditions $u(\lambda_k) \to \bar{u}$ with $\bar{u}_i = 0$ for some $i < n$. Let $I := \{i < n : \bar{u}_i = 0\}$. By Lemma 2, $u_j \leq 0$ for all $j < n$ and all λ, so $\bar{u}_j \leq 0$ for all $j < n$.

Consider $u_i(\lambda_k)/\lambda_k$ for $i \in I$. Either:

Subcase (a): $u_i(\lambda_k)/\lambda_k$ is unbounded below for some $i \in I$. Then $u_i(\lambda_{k_j})/\lambda_{k_j} \to -\infty$ along some subsequence. The bounds from Step 2 give $p^i_{\beta,\lambda}(u) \leq e^{\beta u_i(\lambda_{k_j})/\lambda_{k_j}} \to 0$. The arguments from Steps 2–3 apply, and the normalized first-order condition converges to $v_i = \underline{\beta} \cdot \bar{u}_i$. This gives $\bar{u}_i = v_i/\underline{\beta} \neq 0$ contradicting $\bar{u}_i = 0$.

Subcase (b): $u_i(\lambda_k)/\lambda_k$ is bounded below for all $i \in I$. Then $u_i(\lambda_k)/\lambda_k \in [-M, 0]$ for some $M > 0$. Extract a subsequence along which $u_i(\lambda_{k_j})/\lambda_{k_j} \to c_i \in [-M, 0]$ for all $i \in I$. For $i \notin I$, we have $\bar{u}_i < 0$, so $u_i/\lambda_k \to -\infty$, hence $p^i_{\beta,\lambda}(u) \to 0$.

Rewrite the first-order condition (25) for $i \in I$, multiplied by λ:

$$\sum_{j \neq i} \int \beta \, p^i_{\beta,\lambda}(u) p^j_{\beta,\lambda}(u) \Big[(v_i - \beta u_i) - (v_j - \beta u_j) \Big] dF(\beta) = 0. \tag{26}$$

The integrand is uniformly bounded: $\beta \leq 1$, probabilities lie in $[0, 1]$, and $(v_i - \beta u_i) - (v_j - \beta u_j)$ is bounded since u_i, u_j lie in a compact set. By dominated convergence, we can pass to the limit under the integral. As $\lambda_{k_j} \to 0$: terms with $j \notin I \cup \{n\}$ vanish since $p^j_{\beta,\lambda}(u) \to 0$. For $j \in I \cup \{n\}$, we have $u_j \to 0$, so $(v_i - \beta u_i) - (v_j - \beta u_j) \to v_i - v_j$. The limiting choice probabilities $\bar{p}^i, \bar{p}^j > 0$ for $i, j \in I \cup \{n\}$ are well-defined since $u_m/\lambda \to c_m$ is bounded for $m \in I \cup \{n\}$. Since the left-hand side of (26) equals zero for each λ_{k_j}, the limit must also equal zero:

$$\sum_{j \in I \cup \{n\}, j \neq i} \int \beta \, \bar{p}^i \, \bar{p}^j \, (v_i - v_j) \, dF(\beta) = 0.$$

Let $i^* := \min I$ be the smallest index in I. Since $v_1 < \cdots < v_{n-1}$, we have $v_{i^*} < v_j$ for all $j \in I \cup \{n\}$ with $j \neq i^*$. Thus every term in the sum for $i = i^*$ satisfies $v_{i^*} - v_j < 0$, making the left-hand side strictly negative, which is a contradiction.

Step 5: Convergence. We show that solutions converge to $u^*(v) := v/\underline{\beta}$ as $\lambda \to 0$, uniformly over v in any compact set $V \subseteq \mathbb{R}^{n-1}$ with $v < 0$. Fix such a V. By Lemma 3, solutions lie in a compact set K_V depending only on V.

Consider any sequence (λ^k, v^k) with $\lambda^k \to 0$ and $v^k \to v^* \in V$, and let u^k be a solution at (λ^k, v^k). Since $u^k \in K_V$, there exists a convergent subsequence $u^{k_j} \to \bar{u}$. By Step 4 applied to v^* (the argument applies verbatim to sequences with $v^k \to v^* \in V$), we have $\bar{u}_i < 0$ for all $i < n$. Since $u^{k_j} \to \bar{u}$ with $\bar{u}_i < 0$, the solutions u^{k_j} are bounded away from 0 for large j. The coefficient bounds from Steps 2–3 then apply along this subsequence: the off-diagonal coefficients vanish and the diagonal coefficient converges to $\underline{\beta}$. Taking limits in the normalized first-order conditions yields $v_i^* = \underline{\beta} \cdot \bar{u}_i$, giving $\bar{u} = v^*/\underline{\beta} = u^*(v^*)$. Since every convergent subsequence has the same limit and solutions u^k lies in a compact set, $u^k \to u^*(v^*)$.

For uniform convergence, suppose not: then there exist $\varepsilon > 0$, $\lambda^k \to 0$, $v^k \in V$, and solutions u^k at (λ^k, v^k) with $|u^k - u^*(v^k)| \geq \varepsilon$. Since V is compact, extract a subsequence with $v^{k_j} \to v^* \in V$. By the above, $u^{k_j} \to u^*(v^*)$. By continuity of u^*, $u^*(v^{k_j}) \to u^*(v^*)$. Thus $|u^{k_j} - u^*(v^{k_j})| \to 0$, contradicting $|u^{k_j} - u^*(v^{k_j})| \geq \varepsilon$. □

B. Incomplete Feedback

So far, we have assumed that the principal can choose emotions as an arbitrary function of the action and realized state. In many decision contexts, however, ex post only the payoff is revealed but not the full state. For example, if the agent pursues some economic project, the realized return might be observable but not the return had he pursued another project. In particular, when only the realized material utility $v(a, \omega)$ is learned but not the state ω, the principal cannot in general infer the material utilities of counterfactual actions. Thus, she cannot implement the general regret solution from the previous section. This section generalizes the principal's problem to account for incomplete feedback under the simplifying assumption of a known bias.

Ex-post feedback also plays an important role in the study of regret and disappointment. While classical regret theory (Bell 1982; Loomes and Sugden 1982) does not take into account feedback, several experiments have documented that regret affects decisions more if feedback is expected (Zeelenberg 1999).[22] To accommodate that evidence, Humphrey (2004) introduces a theory of feedback-conditional regret which employs two regret-rejoicing functions, one for complete feedback and one for incomplete feedback. Gabillon (2020) proposes a regret theory for arbitrary feedback structures, where the regret-rejoicing function is applied to the difference between the actual payoff and the highest counterfactual payoff, given the posterior beliefs after feedback. Evidence from Camille et al. (2004) based on self-reported emotions, skin conductance responses, and choices suggests, however, that under partial feedback (only the payoff of the chosen action is learned) primarily disappointment and elation are experienced, while under complete feedback mainly regret and rejoicing are experienced. We are not aware of theories that combine regret and disappointment depending on the feedback structure.

[22] More recently, Somasundaram and Diecidue (2017) find that feedback about foregone options has a polarizing effect on regret attitudes: feedback increases regret aversion among regret-averse decision makers but increases regret seeking among regret-seeking ones.

Our model provides such a theory: we derive optimal emotions as a function of the feedback structure and show that they combine regret and disappointment. To incorporate feedback into our model, we assume that emotions can depend only on what is observed ex post. Formally, we assume the emotion-inclusive utility u is measurable with respect to a feedback structure $\mathcal{F}\colon A \times \Omega \to Z$ where Z is a message space, that is,

$$\forall a, a' \in A, \omega, \omega' \in \Omega\colon \mathcal{F}(a, \omega) = \mathcal{F}(a', \omega') \to u(a, \omega) = u(a', \omega').$$

We consider partitional feedback structures that reveal the material utility $v(a, \omega)$, the action a, and a partition $\mathcal{P}$ of the state space.[23] In one extreme case, the feedback structure reveals the state fully (the finest partition; complete feedback) and in the other extreme, only the material utility is revealed (the coarsest partition; partial feedback). In intermediate cases, some information about the state is revealed, such as general market conditions, while more specific information, such as how each particular economic project would have done, is not revealed.

In some decision-problems, one can learn more about the state from the realized material utility and the action. To ensure that incomplete feedback has bite, we impose a connectedness assumption on the message space. To formalize the condition, we endow the message space Z with a symmetric relation $\sim$, making Z an undirected graph. We define $z \sim z'$ if and only if there exists $\omega \in \Omega$ and $a, a' \in A$ such that $z = \mathcal{F}(a, \omega)$ and $z' = \mathcal{F}(a', \omega)$—that is, two messages are connected if they can be observed under the same state. This is the relevant notion of connectedness because statewise material-utility differences determine incentives for attention, so connected messages must have their emotions determined jointly. The assumption at the end of the following definition rules out cases where messages consistent with a partition cell can be split into disconnected components, which would allow emotions to be chosen independently on each component. A sufficient condition is that any combination of material utilities across actions obtains in some state within the partition cell.

Assumption 5 (Partitional Feedback Structure)**.** *There is a partition $\mathcal{P}$ of Ω such that*

$$\mathcal{F}(a, \omega) = (v(a, \omega), a, \mathcal{P}(\omega))$$

where $\mathcal{P}(\omega)$ is the partition cell that contains ω. We define Z as the image of $\mathcal{F}$ in $\mathbb{R} \times A \times \mathcal{P}$.

For each partition cell $P \in \mathcal{P}$, the subset of the message space that is consistent with this partition cell, $\{(v, a, P) \in Z | v \in \mathbb{R}, a \in A\}$, is a connected subgraph of Z.

For simplicity, we analyze the model under a known bias $\beta \in (0, 1)$. Thus, we consider a pure hidden-action problem without hidden information.

Assumption 6. *The distribution of $\beta \in (0, 1)$ is degenerate.*

[23] A partition of the state space is a set of mutually exclusive and exhaustive subsets $\mathcal{P} = \{P_1, ..., P_n\}$ of Ω. In the Appendix, we show that the resulting emotions are essentially of the same form, when we allow for other deterministic feedback structures that reveal at least the material utility. Further, none of the results below would change if the action was not revealed.

Under known β, we can generalize the model in two dimensions. First, we drop the symmetry assumption on v and the restriction to mechanisms with symmetric u. Second, we generalize the entropy-based attention cost to any posterior-separable cost function. To be coherent with the notation above, we continue to define the attention cost in terms of the state-dependent stochastic choice function.[24] For each state-dependent stochastic choice function $p\colon \Omega \to \Delta(A)$, let $\tau(p) \in \Delta(\Delta(\Omega))$ denote the distribution over posteriors $\gamma \in \Delta(\Omega)$ induced by p via Bayesian updating.

Definition 10 (Posterior-Separable attention cost)**.** *The attention cost c is posterior-separable if*

$$c(p) = \mathbb{E}_{\tau(p)}[H(\mu) - H(\gamma)]$$

where $H\colon \Delta(\Omega) \to \mathbb{R}$ is strictly concave and differentiable on $\Delta(\Omega)$.

Posterior-separable attention costs arise naturally from several foundations, including information theory (Sims 2003), sequential sampling (Morris and Strack 2019; Hébert and Woodford 2023), and constant marginal cost of experimentation (Pomatto, Strack, and Tamuz 2023). The function H can be understood as a measure of uncertainty of a distribution, with entropy as an example. Thus, under a posterior-separable cost, the agent incurs a cost proportional to the expected reduction of the uncertainty of their belief. That H is strictly concave ensures that more information (in the Blackwell sense) is more costly.

Assumption 7. *The attention cost $c(p)$ is posterior separable.*

Given incomplete feedback and the absence of hidden information, the new principal's problem ($\mathrm{P}^{\mathrm{inc}}$) is to pick the solution with minimal socialization cost $\Phi(u)$ to:

$$\max_{u,p} \sum_{\omega\in\Omega}\sum_{a\in A} \mu(\omega)p(a|\omega)v(a,\omega) - \lambda c(p) \tag{$\mathrm{P}^{\mathrm{inc}}$}$$

$$\text{s.t.}$$

$$p \in \arg\max_{\tilde{p}} \beta \sum_{\omega\in\Omega}\sum_{a\in A} \mu(\omega)\tilde{p}(a|\omega)u(a,\omega) - \lambda c(\tilde{p}) \tag{IC}$$

$$\forall a, a' \in A, \omega, \omega' \in \Omega\colon \mathcal{F}(a,\omega) = \mathcal{F}(a',\omega') \to u(a,\omega) = u(a',\omega') \tag{F}$$

We introduce the following class of solutions, which we interpret below as a combination of regret and disappointment. To state it, we define the consideration set $A^*(p) \subseteq A$ of an attention strategy p as the set of actions that have positive probability according to p, formally $A^*(p) = \{a \in A | \sum_{\omega\in\Omega} \mu(\omega)p(a|\omega) > 0\}$.

Definition 11. *A solution (u, p) to the principal's problem* ($\mathrm{P}^{\mathrm{inc}}$) *is a* ***linear regret-disappointment***

[24] This is without loss if the function H below is strictly concave. As is well-known, if we defined the attention cost as a function of the acquired signal structure, we can without loss identify the signal with the action under strictly concave H.

solution *if there exists a function* $\bar{v} : \mathcal{P} \to \mathbb{R}$ *such that for all* $a \in A^*(p), \omega \in \Omega$*:*

$$u(a,\omega) = v(a,\omega) + \left(\frac{1}{\beta} - 1\right)\big(v(a,\omega) - \bar{v}(\mathcal{P}(\omega))\big). \tag{27}$$

The definition states that emotions are a linear function of the difference between the realized material utility $v(a,\omega)$ and a reference point $\bar{v}(\mathcal{P}(\omega))$.[25] The reference point $\bar{v}(\mathcal{P}(\omega))$ of the optimal emotional utility is pinned down by the socialization cost ϕ and depends generally on all material utilities in the partition cell $\mathcal{P}(\omega)$. Thus the emotional utility depends on comparisons to alternative actions (as in regret) and to alternative states (as in disappointment).

For quadratic socialization cost, $\phi(x) = x^2$, the connection to regret and disappointment is more explicit. In this case, Lemma 10 at the end of this section shows that $\bar{v}(\mathcal{P}(\omega))$ is the expected material utility,

$$\bar{v}(\mathcal{P}(\omega)) = \frac{1}{\mu(\mathcal{P}(\omega))} \sum_{\omega' \in \mathcal{P}(\omega)} \sum_{a \in A} \mu(\omega') p(a|\omega') v(a,\omega'). \tag{28}$$

Using this, equation (27) can be rewritten as a weighted sum of comparisons across actions (regret) and across states (disappointment): Let $\gamma \colon A^* \to \Delta(\Omega)$ map each action in the consideration set to its associated posterior under attention strategy p, written $\gamma(\omega|a)$. Conditional on partition cell $\mathcal{P}(\omega)$, define the probability $p(a|\mathcal{P}(\omega))$ and expected material utility $v(a|\mathcal{P}(\omega))$ of action a as

$$\begin{aligned} p(a|\mathcal{P}(\omega)) &:= \sum_{\omega' \in \mathcal{P}(\omega)} \frac{\mu(\omega')}{\mu(\mathcal{P}(\omega))} p(a|\omega') \\ v(a|\mathcal{P}(\omega)) &:= \sum_{\omega' \in \mathcal{P}(\omega)} \frac{\gamma(\omega'|a)}{\gamma(\mathcal{P}(\omega)|a)} v(a,\omega'). \end{aligned}$$

We can then rewrite equation (27) as

$$\begin{aligned} u(a,\omega) = v(a,\omega) &+ \sum_{a' \neq a} p(a'|\mathcal{P}(\omega)) \left(\frac{1}{\beta} - 1\right)\big(v(a,\omega) - v(a'|\mathcal{P}(\omega))\big) \\ &+ p(a|\mathcal{P}(\omega)) \left(\frac{1}{\beta} - 1\right)\big(v(a,\omega) - v(a|\mathcal{P}(\omega))\big). \end{aligned} \tag{29}$$

We interpret the sum in the first line as regret and rejoicing: the weighted difference between the actual material utility and the expected material utilities of all other actions a', conditional on partition cell $\mathcal{P}(\omega)$. The weights are the probabilities of the respective actions under the attention strategy p, conditional on the partition element. This is in contrast to a widely used theory of regret where the comparison is with respect to the ex-post optimal action (Quiggin 1994). That theory is arguably extreme when the agent has a large choice set and does not seem psychologically plausible when the ex-post optimal action was not even considered. Moreover, in some relevant

[25] Linearity arises because the bias β is known and because the bias linearly discounts utilities. As a result, the principal can perfectly offset the bias by scaling up utility differences by $1/\beta$. Under an unknown or nonlinear bias, emotions would generally be nonlinear, as in the main text.

applications as in finance, under no short-sale constraints the ex-post optimal return is infinite (Qin 2020). In contrast, our prediction corresponds to the proposal of Loomes and Sugden (1982) to generalize regret theory to multiple actions based on such a weighted sum of comparisons to *all* other actions, not just the ex-post optimal one. They highlight the importance of providing a theory of the appropriate action weights. Sugden (1986) writes that

> *"Any plausible theory of such weights must, I think, take some account of the extent to which an action is a serious candidate for choice or, to put it slightly differently, of the extent to which the individual could sensibly blame himself for not having chosen it."*

In our context, the action weights are naturally provided by the state-dependent choice probabilities under the attention strategy p. These choice probabilities capture with what probability the agent should have taken some action under the realized state. This implies that regret and rejoicing do not depend on actions outside of the consideration set, which are taken with probability 0. This captures that, as Sugden (1986) writes, only actions that are "serious candidates" or "real options" should be sources of regret and rejoicing. While regret and rejoicing are linear here and thus predict no violations of expected utility axioms from choice, linear regret still has effects on menu choice (Sarver 2008).

We interpret the second line of (29) as disappointment and elation: the difference between the realized material utility $v(a, \omega)$ and the expected material utility of the chosen action, where the expectation is taken with respect to the posterior conditional on the action. Unlike standard models of disappointment and elation (Bell 1985; Loomes and Sugden 1986; Gul 1991), the expected material utility is also conditioned on what is learned about the state (the partition cell). The weight on disappointment and elation is the probability of the actual action conditional on the partition cell.

Proposition 5. *Under Assumptions 5 to 7, any solution of* ($\mathrm{P}^{\mathrm{inc}}$) *is a linear regret-disappointment solution and induces a first-best attention strategy.*

Because we assumed that the bias is known, the optimal mechanism can perfectly offset it by scaling up the utility differences by $1/\beta$, which is achieved by a linear regret-disappointment solution. Using a result in Whitmeyer and Zhang (2022), which builds on the Lagrangian Lemma of Caplin, Dean, and Leahy (2022), we show that such inflating of the utility differences is also *necessary*, in order to implement a first-best attention strategy. The remaining goal of the principal is to reduce the socialization cost of emotions, while scaling up the utility differences. This goal is constrained by the feedback structure. Under complete feedback, the second line of (29) is 0, and only regret is used. However, for incomplete feedback structures, both lines of (29) are generally non-zero and accordingly the optimal emotions can be interpreted as a combination of regret and disappointment.

Disappointment and elation are used because, under incomplete feedback, the emotional utilities for all messages within a partition element are interrelated (Lemma 9). Thus, all material utilities under that partition element, including the material utilities of the chosen action *had another state realized*, affect the optimal emotion under some feedback signal.[26]

Thus, in our model, disappointment and elation can be thought of as *second-best* motivators for attention when complete feedback is not available. They are second-best because they lead to a higher socialization cost than using regret and rejoicing.[27]

One might be surprised that disappointment and elation motivate more attention because they depend on the realization of the state, which the agent cannot affect. By contrast, regret and rejoicing seem to depend more directly on the agent's action, which perhaps more intuitively incentivizes attention. However, although the agent cannot choose the unconditional distribution of states, he can implicitly choose the distribution of states *conditional* on an action by paying attention. Indeed, this is the perspective of attention taken by the posterior approach, in which the agent acquires posteriors over states conditional on each taken action (Caplin and Dean 2013). Like regret and rejoicing, disappointment and elation penalize bad outcomes and reward good outcomes and thus raise the return to attention. As argued above, regret and rejoicing raise the return to attention in the optimal way but under constraints through the feedback structure, disappointment and elation have a role to play.

Proof of Proposition 5. Because the feedback structure reveals the material utility by Assumption 5, the principal can choose the utility $u(z)$ for each message $z = (v, a, P) \in Z$. For $z = (v, a, P)$, we write $v(z)$ and $a(z)$ for the first and third component, respectively. The following lemma uses the fact that the optimal attention strategy under posterior-separable attention costs depends on the statewise utility differences as shown in Whitmeyer and Zhang (2022).

Lemma 9. *A first-best attention strategy p is incentive-compatible under u only if for all elements $z_1, z_2 \in Z$ with $z_1 \sim z_2$ and $a(z_1), a(z_2) \in A^*(p)$, we have*

$$u(z_1) - u(z_2) = \frac{1}{\beta}(v(z_1) - v(z_2)).$$

Proof. By Theorem 4.3 in Whitmeyer and Zhang (2022), to induce an attention strategy that is optimal for the principal, the relative incentives, such as $u(a, \omega) - u(a_1, \omega)$ for $a \in A^*(p), \omega \in \Omega$, are uniquely pinned down. Since the agent discounts the relative incentives by β, for the relative incentives to equal those of the principal, they need to be scaled up by $1/\beta$. This is achieved if and only if for all $\omega \in \Omega$, $u(a, \omega) - u(a_1, \omega) = (v(a, \omega) - v(a_1, \omega))/\beta$. This in turn is equivalent to all elements $z_1, z_2 \in Z$ that are directly connected by an edge (that is, are achieved at the same state) to satisfy $u(z_1) - u(z_2) = (v(z_1) - v(z_2))/\beta$. □

The above is an only-if statement. It ensures only that a first-best attention strategy p would be incentive-compatible if we restricted the agent's choice to the consideration set $A^*(p)$. To implement

[26] This statement holds also under unknown bias.

[27] Under an unknown bias, as we have shown under complete feedback, regret and rejoicing are not only optimal because they minimize the socialization cost but also because they allow tailoring the incentives to the state.

a first-best attention strategy p when the whole action set A is feasible, it is sufficient to punish the actions in $A \setminus A^*(p)$ with sufficiently negative emotions. As long as the principal deters the agent from choosing any action in $A \setminus A^*(p)$, she is indifferent with regards to how strongly she punishes these actions as it does not affect her utility nor the socialization cost.

The utility difference being scaled up by $1/\beta$ is a transitive relation that extends to all nodes on the largest connected subgraph. If z and $\tilde{z}$ are on the same connected subgraph of Z, then there is a finite chain of elements $(z_1, ..., z_k)$ of Z such that $z = z_1$, $\tilde{z} = z_k$, and each adjacent pair is directly connected. If each of those directly connected pairs has a utility difference scaled up by $1/\beta$, then

$$\begin{aligned} u(\tilde{z}) - u(z) &= (u(z_k) - u(z_{k-1})) + ... + (u(z_2) - u(z_1)) \\ &= \frac{1}{\beta}(v(z_k) - v(z_{k-1}) + ... + \frac{1}{\beta}(v(z_2) - v(z_1)) = \frac{1}{\beta}(v(\tilde{z}) - v(z)). \end{aligned}$$

Conversely, if the utility difference of all pairs of nodes on the same connected subgraph is scaled up by $1/\beta$, then trivially the utility difference of directly connected nodes is scaled up by $1/\beta$.

Thus, an optimal attention strategy of the principal is implemented only if the utility difference of all elements on the same connected subgraph are scaled up by $1/\beta$. By Assumption 5, the sets of all messages that are consistent with some partition element $P \in \mathcal{P}$ are connected. By the definition of edges, such sets for different partition elements are not directly connected. In other words, Z has one connected subgraph for each partition element $P \in \mathcal{P}$. This pins down the utility function u except for one constant per subgraph, which is determined by the socialization-cost minimization problem. More specifically, by the above, we can write

$$m(a, \omega) = \left(\frac{1}{\beta} - 1\right) \big(v(a, \omega) - \bar{v}(\mathcal{P}(\omega))\big). \tag{30}$$

The remaining problem is to determine the "reference point" $\bar{v}(\mathcal{P}(\omega))$ for each partition cell. The optimal reference point is connected to the concept of central tendency from statistics. The central tendency $\mathcal{C}_f(X)$ of random variable X with respect to a strictly convex function f is defined as

$$\mathcal{C}_f(X) := \arg\min_c \mathbb{E}[f(X - c)],$$

which is known to be unique under strict convexity of f. Here, we consider the central tendency $\mathcal{C}_\phi$ with respect to the strictly convex socialization cost ϕ.

Lemma 10. *Under socialization cost $\phi(x) = |x|^k$ with $k > 1$, for all $P \in \mathcal{P}$, the socialization-cost minimizing solution to (*$\mathrm{P}^{\mathrm{inc}}$*) has reference point $\bar{v}(P)$ equal to the central tendency of material utility v on the partition element P with respect to an optimal attention strategy,*

$$\bar{v}(P) = \arg\min_{\hat{v}} \sum_{\omega' \in P} \sum_{a \in A} \mu(\omega') p(a|\omega') \phi\big(v(a, \omega') - \hat{v}\big).$$

When $k = 2$, then $\bar{v}(P)$ is the expected material utility,

$$\bar{v}(P) = \frac{1}{\mu(P)} \sum_{\omega' \in P} \sum_{a \in A} \mu(\omega') p(a|\omega') v(a, \omega').$$

Proof. First, the centrality measure of emotions must be 0, otherwise we could decrease the socialization cost. Let $P \in \mathcal{P}$. The attention strategy that is implemented defines a distribution over emotions. The socialization cost on this partition is the expectation of $\phi(u(a,\omega) - v(a,\omega))$. Under the optimal emotions, $\phi(u(a,\omega) - v(a,\omega) - c)$ is minimized at $c = 0$, otherwise one could further reduce the socialization cost. This implies by definition that the central tendency of emotions is 0.

Next, we show two properties of the central tendency. First, $\mathcal{C}_\phi(X + b) = \mathcal{C}_\phi(X) + b$ due to

$$\arg\min_c \mathbb{E}[\phi(X + b - c)] = b + \arg\min_c \mathbb{E}[\phi(X - c)].$$

Second, under $\phi(x) = |x|^k$ with $k > 1$, $\mathcal{C}_\phi(aX) = a\mathcal{C}_\phi(X)$ for all $a \in \mathbb{R}$. First of all, note that $\phi(ax) = |ax|^k = (|a||x|)^k = |a|^k |x|^k = |a|^k \phi(x)$. Then,

$$\arg\min_c \mathbb{E}[\phi(aX - c)] = a \arg\min_c \mathbb{E}[\phi(a(X - c))] = a \arg\min_c \mathbb{E}[|a|^k \phi(X - c)] =$$
$$a \arg\min_c \left(|a|^k \mathbb{E}[\phi(X - c)] \right) = a \arg\min_c \mathbb{E}[\phi(X - c)].$$

Finally, using the two previous facts we show that the reference point of P is the central tendency of the material utility v, with respect to the distribution induced by the implemented attention strategy on P. The reference point $\bar{v}(\mathcal{P}(\omega))$ is the (hypothetical) material utility $\tilde{v}$ that obtains emotions 0 according to equation (30). By equation (30), emotions are a linear function of the material utility $v(a,\omega)$. Then, the realized material utility is also a linear function f of emotions. The material utility $\tilde{v}$ is thus the image of 0 under f, $\tilde{v} = f(0)$. By the above, $\mathcal{C}_\phi(aX + b) = a\mathcal{C}_\phi(X) + b$, so $\mathcal{C}_\phi$ commutes with linear functions. By 0 being the central tendency of emotions, $\tilde{v}$ is the central tendency of the material utility.

As is known, the central tendency $C_\phi(X)$ of random variable X with respect to $\phi(x) = x^2$ equals the expectation of X. □

This concludes the proof of Proposition 5 and of (28). □

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